Pro Football Sports Science Research Update

March 2026

A new conceptual framework for managing hamstring injury risk in soccer – Implementing a data-informed approach: A narrative review. Biology of Sport, 43: 329-353, 2026.

Purpose of the Study

This narrative review aims to propose a new conceptual framework for managing hamstring injury risk in soccer through a data-informed approach. The authors highlight the multifactorial and complex nature of hamstring injuries, particularly those associated with high-speed running (HSR) and sprinting, and emphasize the limitations of current research in establishing clear cause-and-effect relationships. The framework integrates training load (TL) monitoring with biomechanical, neuromuscular, and cognitive factors to support professionals in developing more effective injury prevention and training strategies. While focused on soccer, the review stresses that TL and sprinting are not sole causes but relevant within a broader multifactorial model, advocating for cautious, evidence-based decision-making to mitigate risk without over-relying on simplistic metrics.

Key Findings

- **Multifactorial Etiology of Hamstring Injuries**: Hamstring strains are predominantly non-contact, occurring during HSR or sprinting phases, with risk influenced by a combination of factors including mechanical load exposure (e.g., acute spikes in high-intensity efforts), muscle tolerance to load (e.g., inadequate chronic exposure leading to maladaptation), neuromuscular fatigue, previous injury history, older age, and biomechanical imbalances. The review notes that injuries often happen in the late swing phase of sprinting, involving high eccentric forces on the biceps femoris long head (BFlh).

- **Training Load (TL) Dynamics**: Acute:chronic workload ratio (ACWR) is discussed as a tool, but with caveats—high acute spikes increase risk, while low chronic loads reduce tissue resilience. However, correlations are not always causal, and individual variability is high. GPS-derived metrics like HSR distance (>20 km/h) and sprinting (>25 km/h) are linked to injury, but the evidence is inconsistent due to confounding variables.

- **Cognitive Biases in Injury Causation**: The framework introduces two systems of cognitive bias—fundamental heuristic (System 1: fast, intuitive judgments leading to over-attribution of TL to injuries) and fundamental analytic (System 2: deliberate analysis to avoid biases). Common

biases include assuming TL causation without considering confounders like fatigue or context, potentially leading to suboptimal decisions.

- **Non-Contact Injury Framework**: A model distinguishes adaptation (low risk) from maladaptation (high risk) based on mechanical load exposure and muscle tolerance. It incorporates cognitive biases affecting how practitioners interpret data, emphasizing that while TL metrics can predict risk, they must be contextualized with athlete-specific factors.
- **Evidence Gaps and Limitations**: Much research is associative rather than causal, with inconsistencies in defining TL metrics (e.g., ACWR thresholds vary). Studies on soccer players show higher injury rates in matches vs. training, but applicability is limited by small samples and lack of longitudinal data.

Practical Application for NFL Teams

For sports medicine and performance professionals in the NFL, where hamstring injuries are similarly prevalent (accounting for ~12-16% of injuries, often during sprints or rapid decelerations in plays like routes or tackles), this framework can be adapted to enhance data-driven risk management. Implement GPS or wearable tech to monitor acute load spikes (e.g., avoid >1.3 ACWR) and build chronic loads progressively to improve muscle tolerance—similar to soccer's HSR metrics, track sprint volumes in practices and games. Incorporate cognitive bias checks in decision-making processes, such as multidisciplinary reviews (e.g., involving coaches, trainers, and analysts) to avoid heuristic over-attribution of injuries to "overtraining" without evidence. Use individualized profiles to tailor programs, factoring in player age, injury history, and neuromuscular screening (e.g., eccentric strength tests). This could reduce time lost to hamstrings by focusing on balanced workloads, recovery protocols, and education on biases, ultimately supporting return-to-play decisions and long-term athlete durability in high-stakes NFL environments.

Acceleration and deceleration profiles: Comparison between the 5-0-5 test and seasonal peak player performance. Sports, 14: 9, 2026.

Purpose of the Study

This study, conducted on 19 under-23 Portuguese national-level soccer players, aimed to: (1) assess the intra-session repeatability of the 5-0-5 change-of-direction (COD) test using GPS-derived metrics for acceleration (ACC) and deceleration (DEC); and (2) compare the ACC and DEC profiles from the

preseason 5-0-5 test (including standing-start ACC [ACC_S], DEC prior to the 180° turn [DEC_COD], and re-acceleration after the turn [ACC_COD]) to peak in-season performance values (ACC_max and DEC_max, calculated as the average of the three highest seasonal efforts from training or matches). The goal was to determine if this test could reliably isolate and reflect maximal locomotor capacities in a controlled setting, addressing the challenges of capturing true peaks amid the chaotic nature of games.

Key Findings

- **Repeatability**: Normalized metrics (expressed as a percentage of seasonal maxima) showed good repeatability (ICC = 0.76–0.85), particularly for % Maximum in ACC_COD (ICC = 0.85) and DEC_COD (ICC = 0.76). Absolute values had more variability, with ACC_S showing moderate-to-good repeatability (ICC = 0.73–0.81), but DEC_COD and ACC_COD exhibited poor repeatability for some parameters (e.g., starting speed, ICC = -1.09 to 0.48), likely due to individual movement strategy differences. Low coefficients of variation (CV ~5%) indicated consistent performance across repetitions.
- **Comparison to In-Season Peaks**: The best ACC_COD (5.04 ± 0.39 m/s^2) and DEC_COD (-6.69 ± 0.36 m/s^2) from the test were not significantly different from seasonal ACC_max (5.36 ± 0.54 m/s^2) and DEC_max (-6.35 ± 0.61 m/s^2; $p > 0.05$), reflecting 95% and 107% of seasonal maxima, respectively. However, ACC_S was significantly lower than ACC_max (average: 4.75 ± 0.40 m/s^2, $p = 0.004$, $d = 1.24$; best: 4.88 ± 0.31 m/s^2, $p = 0.004$, $d = 0.98$). Moderate correlations were found between average/best ACC_S and ACC_max ($r = 0.48–0.50$, $p < 0.05$), but no significant correlations for DEC_COD or ACC_COD with seasonal maxima.
- **Additional Insights**: Players covered ~12.1 m accelerating, 2.7 m decelerating, and 5.9 m re-accelerating in the test. Re-acceleration after the turn (ACC_COD) often exceeded initial standing-start ACC, suggesting sport-specific adaptations to COD demands. DEC values in the test surpassed in-season peaks, while most seasonal maxima occurred in matches (Phase 1) rather than training.

Practical Applications

For NFL sports medicine and performance staff, the 5-0-5 test (similar to the pro-agility shuttle) offers a time-efficient, field-based tool to benchmark maximal ACC/DEC capacities during preseason, especially for rookies or players without prior data. Integrate GPS or LP to derive detailed profiles beyond total time, normalizing values to seasonal peaks for reliable monitoring and reducing variability from absolute thresholds (e.g., >3 m/s^2 for high-intensity ACC). This enables individualized training prescriptions—e.g., targeting DEC for positions with frequent sharp cuts (like WRs or DBs)—to optimize load, prevent

underestimation of capacities, and tailor rehab or conditioning programs. Avoid over-relying on test DEC for game simulation, as it may exceed match demands; instead, use it to identify players needing enhanced linear sprint work for overall ACC_max. Limitations include GPS accuracy in sharp COD and small sample size, so validate with your team's tech and larger cohorts.

Association between sports injuries and sleep as measured by sleep actigraphy: A scoping review. The Physician and Sportsmedicine, Ahead of Print: 1-28, 2026.

Purpose of the Study

This scoping review aimed to objectively assess whether sleep patterns—measured exclusively via wrist actigraphy (e.g., using accelerometers in wearables like smartwatches)—are associated with sports-related injuries or athletic performance. The focus was on filling a gap in the literature by relying on actigraphy as a non-invasive, objective alternative to subjective sleep diaries or lab-based polysomnography. Studies on concussions were excluded to emphasize musculoskeletal injuries, and the review included athletes across ages and sports, with data extracted from 17 peer-reviewed articles published in the last 24 years.

Key Findings

The review analyzed 455 athletes (23% female, 77% male) from 10 sports, including football, soccer, rugby, cycling, triathlon, track and field, rowing, tennis, Paralympics, and mixed martial arts. Sleep metrics evaluated included total sleep time/duration, sleep latency (time to fall asleep), wake after sleep onset (WASO; minutes awake after initial sleep), and sleep efficiency (percentage of time in bed spent asleep).

- **Association with Injuries:** Evidence was inconclusive, particularly for adult athletes. Of seven studies, four (57%) reported that poor sleep metrics correlated with higher injury rates, severity, or time missed from play—specifically in elite soccer players (lower sleep efficiency and higher WASO linked to more injuries), Paralympic athletes (longer sleep latency associated with injuries/illnesses), and adolescent track and field athletes (higher WASO as a predictor of recent injuries). However, three studies (43%) found no significant link, including two relevant to football: Burke et al. (2020) on collegiate American football players (no association between sleep duration and injuries) and Dennis et al. (2016) on Australian Football League players (no

difference in sleep duration or efficiency during injury weeks vs. prior weeks). Overall, no consensus emerged for adults, though adolescent data suggested a potential risk.

- **Association with Performance:** Stronger evidence here—all eight studies (100%) linked poor sleep to decreased athletic performance, though impacts varied by type. Poor sleep metrics consistently worsened aerobic performance (e.g., slower race/trial times in cyclists and triathletes, reduced VO2 max and heart rate recovery in rugby players and physically active males, decreased endurance in MMA fighters). Reaction time also declined (e.g., in cyclists and Dutch elite youth athletes). Anaerobic performance showed mixed results: some studies found impairments (e.g., reduced max vertical jump, sled push, or cycling power in MMA fighters and cyclists), while others reported no effect on sprints, strength (e.g., squats, bench press), or sport-specific accuracy (e.g., tennis shots). Additional factors like depth perception (worse in soccer players) and recovery rates (slower in rowers) were negatively affected.

In summary, while injury risk remains unclear (especially in contact sports like football), poor sleep reliably hampers performance metrics critical to NFL play, such as endurance, reaction time, and recovery.

Practical Application

For NFL sports medicine and performance staff, this review underscores the value of prioritizing sleep monitoring and optimization, particularly for performance gains rather than injury prevention (where evidence is weaker). Implement actigraphy via team-issued wearables to track sleep metrics longitudinally—it's practical, cost-effective, and less disruptive than other methods, allowing real-time adjustments during travel-heavy schedules or high-stress periods like playoffs. Encourage players to aim for 7-9 hours of quality sleep nightly, addressing latency and efficiency through routines like consistent bedtimes, reduced screen time, or recovery naps. Given the review's inclusion of football-specific studies showing no injury link, focus interventions on performance: poor sleep could exacerbate fatigue in games requiring sustained aerobic output (e.g., long drives) or quick reactions (e.g., pass coverage). Consider individual differences—e.g., females may need tailored approaches due to hormonal factors—and integrate with existing protocols like load management. Further team-specific studies using standardized actigraphy could clarify injury risks in pro football, but in the meantime, promoting high-quality sleep habits supports healthier, more competitive athletes.

Pro Football Sports Science Research Update

Associations between skeletal muscle mass, fat-free mass, and saliva osmolality: A cross-sectional study. Clinical Journal of Sport Medicine, Ahead of Print: 1-6, 2026.

Purpose of the Study

This cross-sectional study aimed to examine the relationships between skeletal muscle mass (SMM), fat-free mass (FFM), and hydration status as measured by saliva osmolality (SOSM) in a sample of 72 adults (42 men, 30 women; average age ~29 years, body fat ~29%). The hypothesis was that higher SMM and FFM would positively correlate with SOSM, similar to established associations with urinary hydration markers like urine specific gravity (USG) and urine osmolality. The research sought to determine if SOSM, a noninvasive and portable hydration assessment tool, is influenced by body composition in the same way as urine-based methods, which could inform adjustments to hydration guidelines for athletes with varying muscle mass.

Key Findings

- No significant associations were found between SMM or FFM and SOSM. Unadjusted linear regression showed nonsignificant positive trends (SMM: B = 0.44, P = 0.144; FFM: B = 0.23, P = 0.172), which remained insignificant after adjusting for covariates like age, sex, recent fluid intake, and time of day (SMM: B = 0.98, P = 0.146; FFM: B = 0.44, P = 0.173).
- Spearman correlations were small and nonsignificant (SMM and SOSM: r = 0.10, P = 0.427; FFM and SOSM: r = 0.08, P = 0.487).
- Median SOSM was 63.0 mOsm (indicating most participants were hydrated or mildly dehydrated), with no differences based on time of day or fluid intake.
- The lack of association may be due to lower concentrations of muscle metabolites (e.g., creatinine, urea, uric acid) in saliva compared to urine, reducing the influence of muscle mass on SOSM readings.
- In contrast to urine markers, where higher SMM/FFM often leads to elevated readings and potential false positives for dehydration, SOSM appears less biased by body composition.

Practical Applications

For NFL sports medicine and performance professionals, this study highlights SOSM as a promising tool for hydration monitoring, particularly in muscular athletes like linemen or linebackers who may have

high SMM/FFM. Unlike urine-based tests (e.g., USG), which can overestimate dehydration in larger players due to muscle metabolite effects, SOSM shows minimal bias from body composition—potentially reducing misdiagnoses and unnecessary interventions. The MX3 Hydration Testing System used here is portable, noninvasive, and quick (requiring just a saliva sample after a 5-minute fluid abstinence), making it ideal for sideline or locker room assessments during practices, games, or training camps. Teams could integrate SOSM alongside body mass changes or urine tests for more accurate, individualized hydration strategies, especially in hot environments where thermoregulation is critical. However, further validation in elite athletes (e.g., comparing SOSM to gold-standard methods like plasma osmolality) is recommended, and limitations like recent fluid intake should be controlled to ensure reliability.

Beyond the Yard Line: Accommodating rounded sports data in statistical models. The American Statistician, Ahead of Print: 1-24, 2025.

Purpose of the Study

This research addresses a critical but often overlooked issue in NFL analytics: the unique rounding mechanism used to record rushing yardage (and similar metrics like passing/receiving yards). Yardage is technically continuous but officially recorded as integers via a non-standard process—subtracting the floored (rounded down) downing position from the floored line of scrimmage—introducing measurement error, discretization, and bias. The study focuses on running backs' play-level data from the 2023 NFL season, highlighting additional challenges like negative values (e.g., losses behind the line) and right-skewed distributions due to game design and biomechanics. The primary goal is to develop and validate statistical models that correctly incorporate this rounding, using generative distributions for true (unrounded) yardage, to improve accuracy in player performance analysis without access to precise field measurements.

Key Findings

- **Modeling Approach and Superiority of Skewed Distributions**: The authors propose a framework transforming continuous base distributions (normal or asymmetric Laplace distribution [ALD]) into rounded discrete models via convolution and Gauss-Legendre quadrature. In single-player analysis (e.g., Bijan Robinson's 193 carries), the rounded-ALD model outperformed the rounded-normal, showing better fit (DIC: 1134.6 vs. 1247.6) and posterior predictive checks for skewness and kurtosis. It provided more precise estimates (halving

posterior standard deviations) and revealed predictors like "game seconds remaining" (more yards per carry late in games) that were undetectable otherwise.

- **Multi-Player Insights**: Extending to 133 running backs (10,223 carries), a hierarchical rounded-ALD model with random intercepts again excelled (DIC: 57,780.5 vs. 62,399.7). It identified more significant covariates: positive effects from age (peaking mid-career), down number (quadratic), yards to go, line of scrimmage, and opponent rush EPA; negative effects from score differential (more yards when trailing) and game seconds remaining (more yards early). Player rankings shifted notably—e.g., Christian McCaffrey dropped from 2nd to 17th in random intercept rankings due to the model's handling of extreme outliers via skewness—emphasizing how ignoring rounding and asymmetry distorts evaluations.
- **Broader Implications**: Standard continuous treatments of rounded data lead to inefficiency, missed predictors, and non-generative models. The approach confirms rounding's impact propagates through negativity and skewness, affecting conclusions in high-stakes analyses.

Practical Application for NFL Sports Medicine and Performance Professionals

For NFL teams, adopting these rounded models enhances player evaluation by aligning analytics with the true data-generating process, leading to more reliable predictions of rushing performance. This is crucial for injury risk assessment (e.g., factoring fatigue from days rest or travel distance, though not significant here), age-related decline curves (positive age effect with mild quadratic peak, aiding contract decisions), and in-game strategy (e.g., leveraging score differential or time remaining for play-calling). Performance pros can use the open-source code (GitHub: akglazer/rounded-sports-data) to integrate with internal datasets, potentially combining with player-tracking tools for higher-resolution insights—though public ball-tracking data remains limited. Extending to passing/receiving could refine holistic offensive metrics, supporting trades, fantasy projections, betting, and recovery protocols by quantifying how biomechanical factors (e.g., skewness from explosive runs) influence workload and injury propensity. Overall, this reduces bias in metrics, enabling data-driven decisions that optimize player health and team success.

Pro Football Sports Science Research Update

Does longer muscle length resistance training cause greater longitudinal growth in humans? A systematic review. Sports Medicine and Health Science, Ahead of Print: 1-9, 2025.

Purpose of the Study

This systematic review aimed to evaluate whether resistance training (RT) performed at longer muscle lengths (LML-RT), such as through lengthened partials or deeper ranges of motion, leads to greater longitudinal muscle growth compared to RT at shorter muscle lengths (SML-RT). Longitudinal growth refers to increases in fascicle length (FL), a proxy for adding sarcomeres in series, which could contribute to overall muscle hypertrophy. The review focused on studies assessing both FL and muscle size (e.g., thickness, cross-sectional area, or volume) in healthy adults, synthesizing evidence to clarify structural adaptations underlying hypertrophy from these training approaches.

Key Findings

The review analyzed 8 studies involving 120 mostly untrained participants (aged ≥18 years), with training durations of 6–15 weeks. Key muscles examined included the vastus lateralis (most common), triceps brachii long head, and tibialis anterior. LML-RT consistently produced greater overall muscle hypertrophy than SML-RT, with larger increases in muscle size (e.g., +30–50% in anatomical cross-sectional area at distal sites in some quadriceps studies). Evidence for greater longitudinal growth was mixed but modestly favored LML-RT, showing larger FL increases (e.g., +20–36% across proximal-to-distal sites in vastus lateralis) in most studies, potentially indicating enhanced sarcomerogenesis. However, no studies directly measured serial sarcomere number, and FL estimates relied on linear extrapolation methods with questionable validity (only one study used more accurate extended field-of-view ultrasound). Study quality was fair to poor, with limitations including small samples, focus on untrained individuals, and lack of eccentric-only protocols, which may amplify FL changes.

Practical Applications

For NFL sports medicine and performance professionals, prioritizing LML-RT—such as emphasizing the stretched position in squats, leg presses, or overhead triceps extensions—could optimize hypertrophy programs, particularly for building muscle size in key lower-body groups like the quadriceps, which support explosive power and injury resilience. This approach may enhance force production across wider ranges of motion, beneficial for positions involving dynamic movements (e.g., linemen or receivers). Integrate LML elements into periodized training while maintaining full ROM for balanced development and injury prevention, as LML-RT may increase initial muscle damage but confer protective

adaptations over time. Monitor athletes with ultrasound for regional hypertrophy, but note that evidence is preliminary in trained populations—consider pilot testing in offseason programs and await further research on elite athletes to refine protocols.

Echocardiography in Athletes: The ever-evolving assessment of physiology versus pathology. Current Cardiology Reports, 28: 12, 2026.

Purpose of the Study

This review article aims to summarize established physiologic cardiac adaptations observed on echocardiography in athletes, while providing an updated perspective on evolving diagnostic criteria for pathologies that can mimic or overlap with exercise-induced cardiac remodeling (EICR). It emphasizes distinguishing benign adaptations from conditions like apical hypertrophic cardiomyopathy (ApHCM), excessive trabeculations (formerly "non-compaction"), and arrhythmogenic cardiomyopathies. The focus is on helping clinicians interpret echocardiograms in the context of athletic training, considering factors such as sport type, position, sex, race, and comorbidities, to avoid misdiagnosis.

Key Findings

- **Left Ventricle (LV) Adaptations and Pathologies**: Endurance training leads to LV dilation (up to 68-70 mm end-diastolic dimension in elite athletes, vs. normal upper limits of 58 mm for males/52 mm for females), while strength training causes hypertrophy (wall thickness 13-15 mm in the "gray zone"; >15 mm suggests pathology). Asymmetrical or apical hypertrophy favors hypertrophic cardiomyopathy (HCM). Racial differences are notable: Black American football non-linemen showed higher rates of concentric LV hypertrophy (24%) than White counterparts (8%) after 3 years of collegiate play, with similar rates in linemen. Updated ApHCM criteria include indexing maximum wall thickness to body size for better accuracy (78% vs. 59% with prior models). Excessive trabeculations (ratio >2:1 trabecular to compact myocardium) are common in athletes (18.3% prevalence) due to high preload but should not be labeled as cardiomyopathy without systolic/diastolic dysfunction. LV ejection fraction (LVEF) may mildly reduce to ~45% in EICR; contractile reserve testing (e.g., >10% LVEF increase with exercise) differentiates from dilated cardiomyopathy (DCM). Global longitudinal strain (GLS) is typically >-17%, and normal diastolic function supports physiology.
- **Right Ventricle (RV) Adaptations and Pathologies**: Primarily dilation in endurance athletes (e.g., rowers, cyclists), with ~50% showing enlargement. Predictors include male sex, age, and training

intensity. Systolic function remains preserved (e.g., fractional area change ≥32%, RV ejection fraction 52-57%). RV:LV ratio <1 is expected in EICR; disproportionate RV dilation suggests arrhythmogenic RV cardiomyopathy (ARVC). Updated 2020 ARVC criteria raise volume cutoffs (>121 mL/m² for men, >112 mL/m² for women; >130 mL/m² for athletes) and include left-sided phenotypes.

- **Atria, Valves, Coronary Arteries, and Aorta**: Atrial enlargement occurs in both strength and endurance athletes, with reduced contraction strain but preserved reservoir/conduit strains; excessive training may increase atrial fibrillation risk via inflammation/fibrosis. Mild valvular regurgitation is common and benign; moderate/severe requires etiology assessment. Coronary origins should be evaluated for anomalies (e.g., inter-arterial course). Aortic dimensions are generally normal, but >40 mm in men/>34 mm in women warrants workup; >42 mm is pathologic.
- **General Insights**: Echocardiography is first-line for screening (non-invasive, cost-effective) but may need MRI/CT for confirmation (e.g., tissue characterization, scar detection). Factors like hypertension in older athletes can blur lines between EICR and pathology.

Practical Application for NFL Sports Medicine and Performance Professionals

For NFL teams, this review underscores the importance of tailored echocardiogram interpretation during pre-participation physicals, annual screenings, or symptom evaluation (e.g., abnormal ECG). Position-specific considerations are key: Linemen (strength-focused) may show more hypertrophy, while skill positions (endurance elements) exhibit dilation—aligning with the article's football-specific racial data on Black vs. White players, which could inform risk stratification and monitoring protocols. Use contractile reserve or strain imaging to confirm EICR vs. pathology, potentially avoiding unnecessary restrictions on play. If gray-zone findings emerge (e.g., 13-15 mm wall thickness or excessive trabeculations), incorporate MRI for definitive assessment, especially in high-stakes scenarios like draft evaluations or return-to-play decisions. Stay updated on diagnostic shifts (e.g., ApHCM indexing, ARVC criteria) to reduce false positives, optimize player health, and minimize career disruptions. Detraining periods (e.g., offseason) with repeat imaging could help differentiate reversible EICR from fixed pathology, supporting performance longevity and injury prevention strategies.

Pro Football Sports Science Research Update

Effects of nightly electronic device restriction and morning bright light on sleep, mood, and performance among elite athletes. Journal of Strength and Conditioning Research, 39(11): 1155-1162, 2025.

Purpose of the Study

This study investigated the effects of a simple, practical intervention—restricting electronic device use for 30 minutes before bedtime combined with 20 minutes of morning bright-light exposure (approximately 10,000 lux)—on sleep parameters, daytime sleepiness, mood, and physical performance in elite athletes over a 2-week period. The goal was to determine if this light regulation strategy could enhance recovery and performance without requiring major lifestyle changes, addressing common issues like irregular schedules and fatigue in high-level competitors. Athletes from various sports (e.g., fencing, swimming, gymnastics) were included, making it relevant to team sports like football where training demands and travel disrupt circadian rhythms.

Key Findings

- **No Impact on Sleep Parameters**: Objective measures (via actigraphy) and subjective reports showed no significant changes in bedtime, wake time, sleep duration, latency, efficiency, or quality in the intervention group (n=12) compared to controls (n=13). This suggests the protocol did not extend or improve sleep itself, possibly because the 30-minute device restriction was too brief to counteract blue light's melatonin-suppressing effects fully.
- **Improvements in Mood and Alertness**: The intervention significantly boosted Profile of Mood States (POMS) vigor scores ($p < 0.001$) and reduced daytime sleepiness trends (Epworth Sleepiness Scale, $p = 0.095$). Fatigue decreased more in the intervention group ($p = 0.088$), likely due to morning light's alerting and antidepressant effects.
- **Enhanced Performance Metrics**: Reaction time improved ($p = 0.016$), as did rebound jump height ($p = 0.001$) and rebound jump index ($p < 0.001$), indicating better neuromuscular coordination and explosive power. Squat jump height showed a trend toward greater improvement ($p = 0.085$), but no changes occurred in isometric grip strength or heart rate during aerobic exercise. These gains occurred without sleep alterations, attributed to heightened alertness from morning light rather than circadian shifts.

- **Baseline Context**: Athletes had irregular sleep schedules (e.g., 1-1.5 hour delays on free days), with actual irregularities larger than self-reported, highlighting the value of objective monitoring.

Overall, the intervention yielded modest but statistically significant benefits in mood and select performance areas, with no adverse effects, in a small sample of elite athletes (mostly international level, ages 21-26).

Practical Application for NFL Teams

For NFL sports medicine and performance staff, this protocol offers a low-burden, evidence-based tool to combat fatigue and irregular schedules common in pro football (e.g., due to travel, night games, or early practices). Implement it during training camp, recovery weeks, or post-travel by providing bright-light devices (e.g., affordable 10,000-lux lamps) for 20-minute morning sessions and enforcing a 30-minute pre-bed device curfew via team apps or reminders. It could enhance player alertness, mood, and explosive movements (relevant for jumps, sprints, and agility in positions like wide receivers or linebackers) without disrupting sleep routines. Monitor via wearables (e.g., actigraphs) rather than self-reports, as the study showed athletes underestimate irregularities. Start with a 2-week trial on a subset of players recovering from jet lag or high-load periods; combine with existing sleep hygiene for amplified effects. Limitations include the small sample and non-football-specific athletes, so track team-specific metrics like 40-yard dash or vertical jump to validate.

Electromyography activity of posterior oblique sling muscles during high-speed running in elite soccer players: Risk factors for hamstring injury. BMC Sports Science, Medicine, and Rehabilitation, 18(1): 74, 2025.

Purpose of the Study

This cross-sectional study aimed to investigate how activation patterns of the posterior oblique sling (POS) muscles—gluteus maximus (GM), hamstrings (including semitendinosus [ST] and biceps femoris [BF]), and contralateral lumbar erector spinae (CLES)—during the prone hip extension (PHE) test influence their electromyographic (EMG) activity and potential hamstring injury risk during high-speed running (20 km/h on a treadmill). Participants were elite male soccer players without recent injuries or range-of-motion limitations, divided into non-delayed (NDG, n=13) and delayed (DG, n=11) groups based on PHE test results, where DG showed delayed GM and hamstring activation relative to CLES. The

hypothesis was that delayed patterns in low-load tasks like PHE would persist in dynamic activities, altering POS coordination and increasing hamstring strain vulnerability.

Key Findings

- **Group Differences in Muscle Activation Patterns**: Time-series EMG analysis revealed phase-specific variations in POS muscle activity during the running cycle (normalized to 101 points, with early stance ~0-10% and late swing ~90-100%). NDG exhibited higher GM activity in early stance (7-10%), higher ST in late swing (95-100%), and higher BF in mid-swing (45-62%). In contrast, DG showed higher GM in forward swing (55-62%), higher ST in mid-swing (45-70%), and higher BF in early stance (0-15%) and late swing (95-100%). No significant differences were found in CLES activity.
- **Muscle Activity Ratios**: BF-to-GM ratio was higher in DG during early stance (7-11%), indicating reduced GM support and potential BF overload. BF-to-CLES ratio was higher in NDG during mid-swing (52-55%), suggesting better synergistic trunk-lower limb coordination in NDG.
- **Implications for Injury Risk**: DG patterns suggest maladaptive coordination, with delayed GM leading to excessive eccentric loading on BF during high-risk phases (late swing/early stance), potentially increasing hamstring strain susceptibility. NDG patterns aligned with efficient force transfer and reduced overload, supporting lower injury risk. Effect sizes were moderate to large (Cohen's d >0.14 in key phases), and no ROM differences between groups confirmed findings were due to neuromuscular factors.

These results build on prior research showing POS delays in PHE predict hamstring injuries, extending it to dynamic running contexts relevant to field sports.

Practical Application

For NFL sports medicine and performance staff, this study highlights the PHE test as a simple, non-invasive screening tool to identify players with delayed POS activation, who may be at higher risk for hamstring strains during sprints (common in football, akin to soccer's high-speed demands). Implement it preseason or post-injury to stratify athletes into NDG/DG equivalents.

To mitigate risks:

- **Targeted Interventions**: Focus on neuromuscular retraining to optimize POS sequencing. Effective exercises include single-leg bridges (emphasize GM pre-activation), band-resisted bridges, and bird-dog variations with contralateral loading to enhance GM-CLES synergy. Add

reactive drills on unstable surfaces or with resistance bands to improve dynamic pelvic stability and intermuscular timing under load.

- **Integration into Programs**: Incorporate into warm-ups, strength sessions, or rehab protocols, progressing from low-load (e.g., PHE-based cues) to high-speed drills. Monitor EMG if available, or use qualitative cues like trunk control during sprints. This could reduce recurrence rates by addressing root causes like BF overload, complementing traditional strength/flexibility work.
- **NFL-Specific Considerations**: Given football's explosive accelerations, prioritize position groups like wide receivers or running backs. Combine with GPS data to track sprint volumes and adjust training loads for DG-like athletes, potentially lowering the 20-30% hamstring injury incidence seen in the league. Future overground testing (vs. treadmill) could refine applications, as it better mimics game conditions.

Epidemiology of hip and groin injuries in sport: A systematic review with meta-analysis and meta-regression of 5914 injuries from over 9 million exposure hours. British Journal of Sports Medicine, Ahead of Print: 1-9, 2026.

Purpose of the Study

This systematic review and meta-analysis synthesized data from 71 prospective studies (primarily in soccer/football, but including other multidirectional sports like rugby, hockey, and Gaelic football) to quantify the epidemiology of hip and groin injuries (HGIs) in competitive athletes. The primary objectives were to determine injury incidence (injuries per 1000 exposure hours), burden (days lost per 1000 exposure hours), prevalence (e.g., seasonal rates), and the proportion of HGIs relative to all injuries. A secondary aim was to explore moderators influencing incidence, such as age, sex, anthropometrics, playing level, and context (e.g., match vs. training). The study emphasized clinical entities (e.g., adductor-related vs. iliopsoas-related) per the Doha agreement to improve diagnostic and preventive precision. It included 5,914 HGIs across over 9 million exposure hours, focusing on time-loss and non-time-loss cases to avoid underestimation.

Key Findings

- **Incidence and Proportion**: Overall HGI incidence was 0.71 per 1000 exposure hours, representing 11% of all injuries across sports. Adductor-related injuries were the most common

(0.41/1000 hours, 6% of all injuries), followed by iliopsoas-related (0.13/1000 hours, 1%). Other entities (inguinal-, pubic-, or hip-related) each accounted for <1%.

- **Burden**: HGIs caused 11.5 days lost per 1000 exposure hours, with adductor-related injuries contributing the most (6 days/1000 hours). This positions HGIs as frequent and moderately burdensome, comparable to hamstring strains but warranting prioritized prevention.
- **Prevalence**: Seasonal prevalence (over ~9 months) was 19%, with adductor-related at 12%. Non-time-loss injuries (e.g., overuse leading to pain without downtime) may underestimate true rates by 70-90%.
- **Moderators**:
  - **Age**: Incidence increased ~5% per additional year (p=0.046), with higher rates in older cohorts.
  - **Sex**: Males had 1.8 times higher incidence than females (95% CI: 1.25-2.7, p=0.004), potentially due to hip/pelvis morphology differences.
  - **Context**: Matches had 6.5 times higher incidence than training (95% CI: 3.9-10.9, p<0.001).
  - **No Significant Impact**: Height, body mass, BMI, data collection year, or playing level (elite vs. amateur) did not moderate incidence after age adjustments.
- The dataset was soccer-dominant (85%), but findings align with multidirectional sports involving acceleration, deceleration, and directional changes. Small study bias was noted but did not alter core estimates.

Practical Applications for NFL Teams

For NFL sports medicine and performance staff, these findings underscore HGIs as a high-priority issue in American football, given similarities in demands (e.g., sprinting, cutting, and contact in games). Prioritize adductor-focused prevention programs, such as eccentric strengthening (e.g., Copenhagen adductor exercises) integrated into warm-ups and strength sessions, as adductors drive ~57% of HGIs. Implement routine groin symptom screening (e.g., using the HAGOS questionnaire) to catch non-time-loss cases early, especially in older players (>25-30 years) and during games/high-intensity drills, where risk spikes. Tailor load management for males, who face nearly double the risk—consider sex-specific norms in return-to-play protocols. Track burden metrics (e.g., days lost per 1,000 snaps/practices) team-wide to benchmark against the 11.5-day average and justify resource allocation for rehab (e.g., progressive

loading to reduce recurrence). Age-specific strategies are key: monitor adolescents during growth spurts for pubic-related issues, and older veterans for reinjury via personalized plans. Overall, this supports a proactive, data-driven approach to minimize downtime, enhance player availability, and reduce financial/team performance impacts in the high-stakes NFL environment.

Exploratory analysis of relationship between trunk and pelvic motion and shoulder motion during throwing in American football quarterbacks. JSES International, Ahead of Print: 1-24, 2026.

Purpose of the Study

This exploratory study aimed to investigate the kinematic relationships between trunk and pelvic movements at stride foot contact (SFC) during the early cocking phase and shoulder positions at maximum external rotation (MER) during the late cocking phase in American football quarterbacks (QBs). Conducted in a non-game, controlled outdoor setting with 11 male QBs (mean age 25.1 years, experience 10.3 years), the research used markerless motion capture (PitchAI™) to analyze throws over 30 yards at maximum effort. The hypothesis was that trunk/pelvic kinematics at SFC influence shoulder kinematics at MER, similar to patterns observed in baseball pitching, to better understand fundamental throwing mechanics and potential injury risks in QBs.

Key Findings

- **Positive Correlations with Shoulder Horizontal Adduction at MER**: Greater trunk rotation toward the non-throwing side ($r_p = 0.779$, $p = 0.005$), pelvic rotation toward the non-throwing side ($r_s = 0.691$, $p = 0.019$), and hip-shoulder separation ($r_p = 0.667$, $p = 0.025$) at SFC were associated with increased shoulder horizontal adduction at MER. This suggests that "opening up" the trunk and pelvis early may pull the shoulder into a more adducted position later in the throw.
- **Negative Correlations with Shoulder Abduction at MER**: Increased trunk ($r_p = -0.635$, $p = 0.036$) and pelvic ($r_s = -0.682$, $p = 0.021$) rotation toward the non-throwing side at SFC were linked to decreased shoulder abduction at MER, indicating that excessive early rotation might limit shoulder elevation.
- No significant correlations were found with shoulder external rotation at MER or trunk flexion at SFC.

- Mean kinematic values included trunk rotation at SFC of 4.4° ± 29.6°, pelvic rotation of 29.7° (median), hip-shoulder separation of 9.8° ± 11.2°, and shoulder abduction at MER of 98.9° ± 4.7°.
- Limitations: Small sample size (n=11), potential measurement errors from markerless system (RMSE >10° for some shoulder angles), lack of kinetic data, and non-game conditions, which may not reflect in-game pressures.

Practical Applications

For NFL sports medicine and performance staff, these findings highlight the importance of trunk and pelvic control in QB throwing mechanics to potentially mitigate shoulder and elbow injury risks (e.g., rotator cuff tears, impingement, UCL strains). Emphasize coaching cues for maintaining hip-shoulder separation at SFC—such as delaying trunk rotation relative to the pelvis—to promote optimal shoulder horizontal adduction and reduce excessive abduction at MER, which could lower joint loads based on baseball analogies. Integrate markerless tools for field-based assessments during drills, focusing on non-game simulations to establish baselines before progressing to pressured scenarios. In rehab/prevention programs, incorporate exercises targeting rotational sequencing (e.g., medicine ball throws, anti-rotation core work) to enhance kinetic chain efficiency, though larger studies are needed to validate in elite QBs under game-like conditions.

Functional and anatomical characteristics of the heel fat pad based on different sports events. Journal of Biomechanics, 196: 113148, 2026.

Purpose of the Study

This research aimed to evaluate how sport-specific loading environments influence the morphology (thickness) and mechanical properties (stiffness and deformation under load) of the heel fat pad in male university athletes. The heel fat pad was divided into two layers: the superficial microchamber (MIC) layer (closer to the skin) and the deeper macrochamber (MAC) layer (near the calcaneus). Participants included athletes from judo (barefoot, high-impact ground contact), American football (shoed, high-impact with cushioning), swimming (low-impact aquatic environment), and a non-athlete control group. Measurements were taken using ultrasonography under varying loads (0%, 50%, and 100% body weight) and shear-wave elastography for stiffness. The goal was to identify adaptations to different loading patterns, which could inform injury risk and performance in sports with repetitive heel stress.

Pro Football Sports Science Research Update

Key Findings

- **Thickness Variations**: At no load (0%), both MIC and MAC layers were thickest in swimmers and thinnest in judo and American football athletes compared to controls. Under load (50% and 100%), differences persisted, with American football players showing greater MAC layer thickness than swimmers at full load, but overall thinner pads than swimmers at lower loads. After body weight correction (accounting for heavier builds in judo and football players), swimmers consistently had thicker pads, suggesting less degeneration from low-impact activity.
- **Deformation (Rate of Change in Thickness)**: Swimmers exhibited the greatest compression in the MIC layer across all loading transitions (0-50%, 50-100%, 0-100%), indicating higher deformability. In the MAC layer, judo and swimmers showed more deformation than American football players (especially 0-50% and 0-100%), with swimmers also outperforming controls. All groups compressed more in early loading phases (0-50%) than later ones (50-100%), but this was most pronounced in swimmers.
- **Stiffness (Elasticity)**: The MIC layer was stiffest in judo players (likely due to barefoot friction and impact) and softest in swimmers, with American football players and controls in between. The MAC layer was stiffest in American football players and softest in judo players, with swimmers showing the highest elasticity overall. These differences highlight sport-specific adaptations: barefoot sports may toughen the superficial layer but soften the deep layer, while shod sports like football may stiffen the deep layer for impact absorption.

Findings were statistically significant ($p < 0.05$) via ANOVA, with high measurement reliability (ICC ≥ 0.89-0.90). No significant age or height differences across groups, but judo and football players were heavier, and swimmers had longer athletic histories.

Practical Application for NFL Sports Medicine and Performance Professionals

In the NFL, where players experience high-impact, repetitive heel loading in cleats (similar to the study's American football group), these findings suggest potential vulnerabilities in heel fat pad integrity that could contribute to common issues like plantar fasciitis, heel bruises, or chronic pain—conditions that sideline players and impact performance. Football players' thinner pads at rest and stiffer deep layers may indicate adaptations to cushioned footwear, but this could reduce natural shock absorption over time, increasing stress on the calcaneus during cuts, tackles, or sprints.

- **Assessment and Monitoring**: Incorporate ultrasound evaluations (e.g., during pre-season physicals or injury evals) to measure heel fat pad thickness and stiffness, especially for linemen

or skill players with high ground-force demands. Track changes over a career to detect early degeneration, using body weight-corrected metrics for accuracy in larger athletes.

- **Injury Prevention**: Design footwear interventions (e.g., custom insoles with variable cushioning) to mimic swimmers' higher deformability, potentially reducing overload. Integrate low-impact cross-training (e.g., pool sessions) to preserve pad thickness and elasticity, as seen in swimmers. For rehab, focus on barefoot or minimal-shoe exercises cautiously to enhance superficial stiffness without over-stressing the deep layer.
- **Performance Optimization**: Thinner, stiffer pads in football players may correlate with reduced energy return or increased fatigue—consider this in load management protocols. Tailor training for rookies (often with less adaptation) vs. veterans, and use these insights to inform return-to-play decisions post-heel injury, aiming to restore pre-injury deformability for better shock absorption.

Overall, this study underscores the need for sport-tailored foot health strategies in the NFL to mitigate long-term wear from shod, high-impact play.

Genetic and epigenetic determinants of injury risk and recovery in elite athletes: Toward precision sports medicine. Gene, 980: 149957, 2026.

Purpose of the Study

This narrative review aims to synthesize evidence on genetic and epigenetic factors influencing musculoskeletal injury risk and recovery efficiency in elite athletes. It focuses on elite performers (defined as those with at least five years of high-performance training and national/international competition experience) and addresses unexplained interindividual variability in injury susceptibility and adaptation. The review evaluates candidate genes, genome-wide association studies (GWAS), polygenic risk scores (PRS), and epigenetic mechanisms (e.g., DNA methylation, microRNAs) as bridges between genetics and environmental factors like training load. It highlights opportunities for precision sports medicine while identifying challenges like ethnic bias, data privacy, and clinical translation.

Key Findings

- **Genetic Determinants of Injury Risk**: Single-nucleotide polymorphisms (SNPs) in connective tissue genes like COL1A1 (rs1800012 TT genotype protective against tendon/ligament injuries, OR=0.17) and COL5A1 (rs12722 TT genotype increases risk, OR=1.58) influence collagen integrity

and are linked to injuries such as ACL tears and tendinopathies. Muscle-related SNPs, including ACTN3 (R577X XX genotype associated with higher muscle damage post-eccentric exercise), CKM (rs8111989 GG genotype favors power athletes), and IL6 (rs1800795 C allele linked to prolonged inflammation), contribute to strains and recovery delays. GWAS in athlete cohorts (e.g., Estonian National Team) identified novel loci like PAPPA2 and COL22A1 with suggestive associations (OR up to 13.8) for specific injuries, emphasizing polygenic architecture over single genes.

- **Epigenetic Mechanisms**: Dynamic factors like DNA methylation and exercise-responsive microRNAs (e.g., miR-206, miR-133a, miR-486) regulate gene expression in response to training, nutrition, and stress. Concepts like "epigenetic muscle memory" show retained hypomethylation of hypertrophy genes (e.g., UBR5, GRIK2) after detraining, enabling faster adaptation upon reloading. Sex- and age-specific effects were noted, with stronger associations in females (e.g., COL1A1 T allele and ACL risk) and older athletes (e.g., IL6 C variant impairing healing).
- **Polygenic and Multi-Omics Integration**: PRS aggregating multiple SNPs (e.g., ACTN3, COL5A1, IL6) improve prediction (e.g., 3.5x higher muscle injury risk below TGS threshold in football players). Multi-omics (genomics, metabolomics, proteomics) reveals signatures like elevated 4-hydroxyproline during recovery, supporting "sportomics" for real-time profiling. Limitations include small cohorts, Caucasian bias, and inconsistent replication across sports.

Practical Applications

For NFL teams, integrating genetic/epigenetic profiling into player management could enable personalized injury prevention and rehab protocols. Use PRS and GWAS data to stratify players by risk (e.g., genotyping for COL5A1/ACTN3 to flag high-risk athletes for hamstring/ACL issues common in football). Combine with wearables and AI for real-time load monitoring—e.g., adjust training for IL6 C carriers prone to inflammation delays. Epigenetic biomarkers (e.g., miR-486 levels) could guide nutrition/recovery timing, optimizing return-to-play for "non-responders." Ethical considerations: Ensure privacy and avoid misuse (e.g., talent selection bias). Start with pilot testing in academy/roster evaluations to refine precision strategies, potentially reducing time-loss injuries and extending careers.

Pro Football Sports Science Research Update

How do muscle injuries relate to return to performance metrics in male elite football players? Biology of Sport, 43: 165-175, 2026.

Purpose of the Study

This longitudinal prospective study examined 333 male elite football (soccer) players from an elite Spanish club over five seasons (2017/18 to 2021/22), focusing on 234 lower-limb muscle injuries (hamstring and quadriceps). The primary aim was to investigate how injury type, location, and severity influence the time required to return to pre-injury performance levels after return to play (RTP). Performance was assessed via GPS-tracked metrics: high-speed running (HSR >21 km/h), sprint distance (>24 km/h), and maximum velocity (V_max). The study emphasized the "return-to-performance" (RTPerf) phase, defined as the period post-RTP needed to achieve or exceed pre-injury fitness, highlighting gaps in prior research that focused mainly on RTP without post-return load analysis.

Key Findings

- **Hamstring Injuries (n=181, including medical attention cases):** These had the most pronounced impact on performance, with severity playing a critical role. Severe injuries (>28 days off) led to significant reductions in V_max (e.g., -1.43 km/h in Post1 week, persisting at -0.90 km/h by Post5) and HSR (-32.90 m in Post1). Mild injuries (1-7 days) showed HSR drops in Post1 (-45.81 m), while moderate injuries (8-28 days) affected V_max in Post1 (-1.00 km/h). Hamstrings accounted for higher injury burden (up to 913 days/1,000 exposure hours for severe cases) and recurrence rates (up to 34% in the same season). Pre-injury, HSR and sprint metrics often dipped in the week before injury, potentially due to fatigue.

- **Quadriceps Injuries (n=67):** These showed less severe and shorter-lasting effects. Mild injuries reduced HSR (-52.70 m) and V_max (-1.18 km/h) in Post1, with recovery typically by Post2. Moderate injuries impacted sprint distance up to Post2 (-43.09 m), but severe cases showed no significant post-injury changes in metrics. Overall, quadriceps recovery was faster, with minimal long-term deficits compared to hamstrings.

- **General Patterns:** Injuries requiring medical attention (0 days off) had minimal impact, but time-loss injuries correlated with greater performance decrements. Linear mixed models confirmed location-specific differences: hamstrings needed up to 5 weeks for full RTPerf, while quadriceps often recovered within 1-2 weeks. No major pre-injury load spikes were noted, but subtle reductions in high-intensity metrics preceded some events.

Pro Football Sports Science Research Update

Practical Application for NFL Sports Medicine and Performance Professionals

This research underscores the need for tailored, location-specific protocols in managing hamstring and quadriceps injuries, which are common in American football due to similar demands for explosive sprints, decelerations, and high-speed actions (e.g., wide receivers or running backs). For hamstrings—often the top time-loss injury in the NFL—implement gradual load progression over 4-5 weeks post-RTP, monitoring GPS metrics like HSR and V_max to avoid reinjury (which mirrors NFL recurrence rates of 20-30%). Use progressive exposure to game-like demands, incorporating neuromuscular testing (e.g., isokinetic strength) and psychological readiness assessments to ensure safe RTPerf.

For quadriceps, focus on early-week monitoring post-RTP, as deficits resolve quicker; mild cases may need just 1-2 weeks of adjusted loads. Coordinate medical/rehab teams with strength coaches for daily load tracking, potentially using GPS and LP systems. This could reduce economic impact (e.g., akin to the study's €500,000/month per player estimate) and boost team success by maximizing availability. Extend follow-ups beyond 5 weeks in NFL contexts, accounting for position-specific demands, and integrate fatigue monitoring to preempt injuries. Overall, these insights promote a "control-chaos continuum" approach, balancing controlled rehab with chaotic game simulations for optimal outcomes.

Income taxes and firm competitiveness: A case study from the National Football League. International Journal of Financial Studies, 13(4): 212, 2025.

Purpose of the Study

This research investigates the impact of state income taxes on NFL team competitiveness, using the league's salary cap (implemented in 1994) as a natural experiment. By comparing NFL teams (affected by taxes due to player salaries) to college football teams (unaffected, as players were amateurs during the 1984-2000 study period), the authors employ a difference-in-differences model to estimate causal effects on performance. The goal is to understand how taxes create distortions under a binding budget constraint like the salary cap, extending economic theory on tax incidence to professional sports. Performance is measured via winning percentage and Simple Rating System (SRS), a metric accounting for point differential and strength of schedule to better capture true team quality.

Key Findings

- **Negative Impact of Taxes on Performance**: Higher state income taxes reduce team quality, with the effect amplified post-salary cap. A one standard deviation increase in tax rate (about 3.4%) is

associated with roughly 0.32 fewer wins per season (though noisy and not always statistically significant for winning percentage). The impact is stronger and more consistent for SRS, showing a decline of 31-55 points in baseline models, worsening by an additional 35-36 points after 1994.

- **Robustness Across Analyses**: Results hold using various control groups—full college football (2,333 observations), power conferences (1,881 observations), and hand-matched in-state pairs (1,187 observations)—with stronger statistical significance in restricted samples.
- **Shift in Competitive Dynamics**: Pre-cap (1984-1993), teams in high-tax states (≥7.2%) were more likely to be above average (60% probability of positive SRS vs. 44% for low-tax states). Post-cap (1994-2000), this reversed: high-tax states (≥7%) had only 36% probability of above-average performance vs. 58% for low-tax states. A secondary split post-cap suggests potential nonlinear effects, where ultra-high taxes (≥7.9%) might correlate with better outcomes (53% above average), possibly due to state investments in infrastructure.
- **Theoretical Insights**: Taxes increase effective player costs for teams (via higher gross salaries to offset player tax burden), especially for inelastic positions like quarterback, leading to roster distortions and reduced overall talent depth.

Practical Application

For sports medicine and performance professionals in NFL teams, particularly those in high-tax states (e.g., California, New York), this highlights how taxes can constrain roster building, potentially leading to shallower depth charts and increased reliance on your expertise to optimize player health and output. Teams may prioritize drafting or trading over free agency to conserve cap space, resulting in younger, less experienced rosters prone to higher injury rates or performance variability—emphasizing the need for robust injury prevention, recovery protocols, and conditioning programs. Advocate for budget allocations toward advanced medical tech, analytics-driven load management, or staff expansions to mitigate these disadvantages. League-wide, findings support policies like compensatory draft picks for high-tax teams losing free agents, which could indirectly enhance resources for player wellness and long-term performance sustainability. Overall, integrate tax awareness into cross-departmental strategies to maximize competitive edge despite fiscal hurdles.

Pro Football Sports Science Research Update

Independent effects of mental fatigue and drop height on drop jump performance in American football athletes: An exploration of central-peripheral interaction. PeerJ, 14: e20574, 2026.

Purpose of the Study

This study aimed to examine the independent effects of mental fatigue (MF) and varying drop heights (30 cm, 40 cm, and 50 cm) on drop jump (DJ) performance in male American football athletes. It specifically explored central-peripheral interactions, using the DJ as a model to simulate the stretch-shortening cycle (SSC) common in football movements. The research addressed a gap in understanding how MF—induced by prolonged cognitive tasks like decision-making under pressure—affects explosive lower-limb performance in high-intensity sports like American football, where athletes face both physical and mental demands. Hypotheses included: (1) MF has less impact on low-height DJs; (2) MF does not significantly affect jump height; and (3) drop height significantly impacts DJ execution. Twelve collegiate athletes (age 22.2 ± 1.9 years, height 181.4 ± 3.1 cm, weight 80.1 ± 8.5 kg) participated in a repeated-measures design, with MF induced via a 45-minute Stroop task.

Key Findings

The study used biomechanical tools (Vicon motion capture, Kistler force platforms) to measure DJ metrics under baseline and MF conditions. A 2 × 3 repeated-measures ANOVA revealed no interaction effects between MF and drop height, indicating independent influences. Key results are summarized below:

Metric	Effect of MF	Effect of Drop Height	Notes
Jump Height (normalized to body height)	No significant effect	Lower at 40 cm (p=0.002) and 50 cm (p=0.008) vs. 30 cm under both conditions	Supports hypothesis 2; athletes absorbed more impact at higher heights, reducing takeoff efficiency.
Peak Vertical Ground Reaction Force (vGRF_peak)	No significant effect	No significant effect	Minimal changes in max force generation.

Pro Football Sports Science Research Update

Metric	Effect of MF	Effect of Drop Height	Notes
Rate of Force Development (RFD)	No significant effect	Lower at 40 cm and 50 cm vs. 30 cm ($p<0.001$) under both conditions	Indicates slower force buildup at higher heights, potentially due to conservative strategies to manage impact.
Average Power Output (W/kg)	Lower under MF across all heights ($p<0.001$)	No significant effect	Suggests MF impairs central drive, reducing overall power.
Reactive Strength Index (RSI, m/s)	Lower under MF across all heights ($p=0.001$)	Lower at 40 cm ($p=0.004$) and 50 cm ($p=0.008$) vs. 30 cm	Highlights reduced reactive efficiency from both MF and higher heights.
Reactive Strength Ratio (RSR)	Lower under MF across all heights ($p=0.001$)	No significant effect	Points to MF's role in prolonging contact time relative to flight time.
Lower Limb Stiffness (K_leg, BW/m)	No significant effect	Higher at 40 cm ($p=0.001$) and 50 cm ($p=0.008$) vs. 30 cm	Increased stiffness at higher heights may reflect greater muscle pre-activation to absorb impact, but at the cost of performance.

MF was successfully induced (increased cognitive demand and effort via NASA-TLX scores). Hypotheses 1 and 2 were partially supported (no differential MF impact by height; no MF effect on jump height), while hypothesis 3 was confirmed (drop height significantly affected multiple metrics).

Practical Applications

For NFL sports medicine and performance professionals, this study underscores the need to address MF as a modifiable factor that can impair central nervous system efficiency, leading to reduced power output, RSI, and RSR—key for explosive movements like tackling, sprinting, and jumping in football. Monitor MF sources (e.g., pre-game decision pressure, smartphone use, sleep deprivation) and incorporate brain endurance training or recovery strategies (e.g., mindfulness, short cognitive breaks) to mitigate declines, especially during high-stakes periods like playoffs. Experienced athletes may build resilience through psychological skills training, helping maintain performance under fatigue.

Pro Football Sports Science Research Update

Regarding drop heights in plyometric training: Favor 30 cm for American football-specific drills, as it aligns with game demands (rapid, multidirectional explosiveness) and optimizes SSC efficiency without excessive impact. Avoid routine use of 40-50 cm heights, which increase leg stiffness and injury risk (e.g., via higher forces on muscles/tendons) while reducing jump height and RFD—potentially counterproductive for agility-focused positions like wide receivers or defensive backs. Tailor heights based on individual strength levels; for stronger athletes, 30 cm may suffice for maintenance, while monitoring for overuse. Integrate these insights into periodized programs to enhance fatigue management, reduce injury rates, and optimize competitive state. Future NFL-applied research could test these in pro athletes under game-like conditions for broader generalizability.

Injectable Peptide Therapy: A primer for orthopaedic and sports medicine physicians. The American Journal of Sports Medicine, 54(1): 223-229, 2026.

Purpose of the Study

This narrative review aims to assess the current scientific evidence on injectable peptide therapies, with a focus on their potential applications in regenerative medicine (e.g., tendon, muscle, and ligament repair) and sports performance enhancement. It targets orthopaedic and sports medicine physicians, providing an overview of popular peptides like BPC-157, TB-4/TB-500, CJC-1295 + ipamorelin, tesamorelin, and GHK-Cu to equip providers with knowledge for patient counseling amid growing market popularity and direct-to-consumer marketing.

Key Findings

- **BPC-157**: Preclinical animal studies show promising effects on tendon-to-bone healing, muscle regeneration, and reduced inflammation via mechanisms like nitric oxide modulation and growth factor upregulation. However, human evidence is limited to one small, methodologically flawed case series (n=12) reporting subjective knee pain improvement after intra-articular injections, with no controls, validated outcomes, or standardized follow-up. No randomized trials exist, and toxicity data are incomplete.
- **TB-4 and TB-500**: Animal models demonstrate benefits in angiogenesis, wound healing, muscle fiber regeneration, and anti-inflammatory effects (e.g., reducing TNF-α). TB-500, a synthetic fragment of TB-4, was originally used in veterinary contexts like horse racing but lacks human studies. Both show tumor-related concerns in some research (e.g., potential metastasis promotion), though conflicting data exist. No orthopaedic clinical trials have been conducted.

Pro Football Sports Science Research Update

- **CJC-1295 + Ipamorelin**: This combination boosts growth hormone (GH) and insulin-like growth factor-1 (IGF-1) levels in animal models, leading to reduced muscle atrophy and improved bone growth in glucocorticoid-induced conditions. Human trials are limited to non-orthopaedic uses (e.g., GH deficiency), with one discontinued due to a cardiac event. No evidence supports musculoskeletal applications.
- **Tesamorelin**: FDA-approved for HIV-associated lipodystrophy, it increases IGF-1 and improves mitochondrial function in obese patients with GH deficiency. No preclinical or clinical data exist for orthopaedic or sports-related uses.
- **GHK-Cu**: Shows antioxidant, anti-inflammatory, and wound-healing effects in animal and in vitro studies, potentially aiding tissue repair via copper metabolism. However, no clinical evidence supports its use for musculoskeletal conditions, and angiogenesis promotion raises theoretical cancer risks.
- **Overall**: While peptides offer theoretical advantages (e.g., high bioavailability, low immunogenicity) over biologics like PRP, there is a profound lack of human clinical trials for orthopaedic indications. No FDA approvals exist for musculoskeletal use; many are categorized as high-risk (503A Category 2) due to safety concerns like impurities. Several (e.g., TB-4/TB-500, tesamorelin) are on the WADA Prohibited List. The global peptide market is booming ($37.8B in 2022, projected to $85.9B by 2031), but marketing often relies on unsubstantiated claims via social media and online sales.

Practical Application for NFL Sports Medicine and Performance Professionals

For NFL teams, this review underscores the need for caution in discussing or managing peptide use among athletes. Given the absence of robust human evidence for efficacy in injury recovery or performance enhancement (e.g., muscle repair or reduced recovery time), peptides should not be recommended or administered. Emphasize WADA/USADA prohibitions—many peptides qualify as banned substances, risking suspensions (e.g., TB-500 has been linked to doping scandals in horse racing and could trigger similar scrutiny in humans). Counsel players on risks, including unknown dosing, potential impurities from unregulated sources, and side effects like inflammation or tumorigenesis. Prioritize evidence-based alternatives like PRP or physical therapy for regenerative needs. Monitor emerging research, as peptides may hold future potential, but current data do not support integration into NFL protocols. If athletes inquire (e.g., due to media hype), document discussions and refer to league anti-doping policies to mitigate liability.

Pro Football Sports Science Research Update

Integrating intramuscular fat radiomics with hamstrings to quadriceps structure and function ratios to predict future hamstring strain injury. PLOS Digital Health, 4(12): e0001144, 2025.

Purpose of the Study

This prospective longitudinal study aimed to evaluate whether MRI-derived radiomic features from intramuscular fat (IMF) in the hamstrings and quadriceps could predict future hamstring strain injuries (HSI) in professional American football players. Additionally, it investigated if integrating these radiomic features with muscle imbalance metrics (e.g., hamstrings-to-quadriceps (H:Q) ratios for cross-sectional area (CSA) and torque) and injury profile data (e.g., prior HSI history and Hamstring Outcome Score (HaOS)) could enhance predictive accuracy. The cohort included 93 NFL players (mean age ~25 years), with MRI scans (using IDEAL sequences) collected preseason, followed by tracking for in-season HSIs. Machine learning (K-nearest neighbor classifier) was used to build and compare models.

Key Findings

- **Radiomics Alone**: Features extracted from hamstring IMF (e.g., texture patterns like gray-level co-occurrence and run-length matrices) predicted future HSI with moderate accuracy (AUC = 0.68; 95% CI: 0.67–0.69). Quadriceps IMF radiomics performed similarly but slightly worse (AUC = 0.69 when integrated with other factors).
- **Integrated Models**: The top-performing model combined hamstring IMF radiomics with muscle imbalance (H:Q CSA and torque ratios) and injury profile (prior HSI and HaOS scores), achieving an AUC of 0.79 (95% CI: 0.78–0.79). This significantly outperformed models using quadriceps radiomics (AUC = 0.69), radiomics alone, imbalance alone (AUC = 0.64), or injury profile alone (AUC = 0.68).
- **Other Insights**: Injured players (n=9) showed lower H:Q torque ratios (0.39 vs. 0.49 in uninjured; p=0.006) and reduced hamstring torque, suggesting functional imbalances contribute to risk. No significant differences in age, weight, height, or CSA were found between groups. The study highlighted IMF's role in post-injury muscle disruption, with radiomics capturing subtle textural changes not visible in standard MRI assessments.
- **Limitations**: Small injured sample size (9/93), all-male elite cohort, and need for external validation in larger, diverse populations (e.g., including females and non-professionals).

Pro Football Sports Science Research Update

Practical Application

For NFL sports medicine and performance teams, this approach offers a preseason screening tool to identify high-risk players for targeted interventions, potentially reducing HSI incidence (which accounts for ~2.2 per 1,000 exposures in training camps and significant time loss/economic burden). Integrate MRI radiomics (focused on hamstring IMF) with simple assessments like isokinetic torque testing and HaOS surveys into routine evaluations. High-risk athletes could receive personalized programs emphasizing hamstring strengthening, imbalance correction, and IMF monitoring to mitigate fatty infiltration. While not yet clinically validated at scale, this could inform return-to-play decisions and prevention strategies, with code available on GitHub for replication. Larger studies are recommended before widespread adoption.

Jones fractures in elite athletes – 2025 International Foot and Ankle Sports Consensus and Systematic Review. Knee Surgery, Sports Traumatology, Arthroscopy, Ahead of Print: 1-15, 2025.

Purpose of the Study

The study aimed to establish an expert consensus on the management of Jones fractures (fractures at the proximal metaphyseal-diaphyseal junction of the fifth metatarsal) in elite athletes, using a modified Delphi technique involving 37 international foot and ankle specialists. This was supported by a systematic review of literature from 2004 to 2024 to compare nonoperative versus operative approaches, focusing on outcomes like return to sport (RTS), complications, and treatment failure. The goal was to provide evidence-based guidance for clinicians, addressing diagnostic workup, conservative and surgical treatments, and RTS considerations, particularly for high-demand athletes where delayed healing due to poor vascularity in the fracture zone can prolong recovery.

Key Findings

- **Consensus Statements:** Out of 37 statements, 15 achieved unanimous consensus (100% agreement), 21 strong consensus (86-99%), and 1 consensus (75-85%). Key areas included:
    - **Diagnosis:** Plain radiographs as first-line imaging (unanimous); MRI for nondisplaced fractures and CT for displaced/comminuted ones (unanimous); evaluate training loads, biomechanics, and systemic factors like vitamin D deficiency (strong to unanimous).

- **Conservative Management (for chronic incomplete fractures):** Indications limited to non-apparent MRI findings (strong); use walking boot over stiff-soled shoe (unanimous); non-weight bearing for 4-6 weeks, progressing to partial at 6 weeks (strong); vitamin D/calcium supplementation recommended (strong); avoid NSAIDs initially (strong); RTS to light training in 2-4 months, full sport in 4-6 months (strong).
- **Surgical Management:** Preferred for displaced fractures, nonunions, conservative failures, or high-level athletes needing early RTS (strong); minimally invasive percutaneous screw fixation for nondisplaced (unanimous); intramedullary screw for displaced (strong); bone grafting for nonunions (unanimous); biologics (e.g., BMAC, PRP) may be used (strong); non-weight bearing for 3-4 weeks post-op, partial at 4-6 weeks, full at 6 weeks (strong to unanimous); start PT at 2 weeks (strong); RTS to light training in 2-3 months, full sport in 3-6 months (unanimous to strong). Bone stimulators not routine but optional (consensus).

- **Systematic Review:** Analyzed 27 studies (2 RCTs, 12 retrospective, 13 case series) with 1056 patients (1070 feet). Fractures classified as stress (chronic, repetitive; n=699) or traumatic (acute; n=357). Mean age: 21.1 years (stress), 29.6 years (traumatic). Top sports: soccer (stress), American football (traumatic).
  - **Stress Fractures:** 86.4% treated with ORIF; mean RTS 13.3 ± 4.5 weeks; complication rate 8.4%. Nonoperative failure rate highest at 8.2%.
  - **Traumatic Fractures:** 62.7% surgical, 37.3% conservative; all returned to sport at 11.6 ± 2.2 weeks regardless of treatment; conservative failure highest at 16.1%.
  - Overall, surgical treatment yielded shorter RTS than nonoperative, regardless of etiology or chronicity, with lower failure rates. American football players represented 4.7% of stress and 28.3% of traumatic cases, often at professional (12-28%) or collegiate (10%) levels.

Practical Application for NFL Teams

For NFL sports medicine and performance staff, this study supports prioritizing surgical intervention (e.g., percutaneous intramedullary screw fixation) for Jones fractures in players, especially those with stress etiology or high performance demands, to minimize downtime—potentially reducing RTS from 4-6 months (conservative) to 3-6 months (surgical), critical during a 17-game season. Early MRI/CT imaging and assessing factors like training load, biomechanics (e.g., custom orthotics), and vitamin D levels can

prevent refractures, which occurred in 8.4% of surgical cases. Rehab protocols should include non-weight bearing for 3-4 weeks post-op, early PT at 2 weeks, and gradual progression to full weight bearing at 6 weeks, incorporating low-impact cross-training. Avoid initial NSAIDs to aid healing, and consider biologics or bone grafting for nonunions. Given American football's prominence in traumatic cases, integrate these into pre-season screening and injury protocols to optimize player availability and long-term foot health.

Kinematic sequencing of the football pass using inertial motion analysis. International Journal of Sports Physical Therapy, 20(8): 1214-1221, 2025.

Purpose of the Study

This pilot descriptive cross-sectional study aimed to analyze the kinematic sequencing of the football pass in quarterbacks using wireless inertial measurement units (IMUs). Specifically, it sought to capture joint angles and motion patterns across the lower extremities, trunk, pelvis, and upper extremities during drop-back passes at varying distances (10, 20, and 30 yards). The research addressed a gap in football-specific biomechanics, where most prior studies focused on baseball pitching, by leveraging portable IMU technology to enable field-based assessments without the constraints of lab-based optical systems.

Key Findings

Eight healthy, right-handed male quarterbacks (four high school, four collegiate; mean age 16.4 ± 1.8 years) performed nine passes while equipped with 16 Noraxon Ultium IMUs. Data were analyzed at four points of interest (POI): stride foot contact, maximal shoulder external rotation (MaxER), ball release, and maximal shoulder internal rotation (MaxIR). Key kinematic insights include:

- **Upper Extremity**: Shoulder abduction peaked at 112° and external rotation at 134° just before acceleration (at MaxER), with elbow flexion reaching 100°. These values declined post-MaxER, with shoulder external rotation dropping to 56° at ball release and elbow extension to 31°.
- **Lower Extremity**: Stride knee flexion remained stable (29°–46°), peaking at ball release, indicating a consistent base for force generation.
- **Trunk and Pelvis**: The trunk stayed relatively upright, with 23° lumbar extension at MaxER. Transverse plane motion was prominent: trunk rotation shifted from 40° right (throwing side) at foot contact to 21° left at MaxIR. Pelvis rotation began earlier in cocking, facing the target with

2°–6° left rotation. Hip-shoulder separation averaged ~20°, favoring right trunk rotation in cocking but reversing post-release, with minimal separation (11°) at ball release—suggesting optimal force transfer when the body aligns near neutral.

- **Overall Sequencing**: From lower to upper body, motion converged to near-neutral positions in all planes before ball release, emphasizing efficient kinetic chain transfer. Compared to baseball pitching, football passes showed less elbow extension at release (121° vs. 150°–160° in baseball) and more stable knee flexion, likely due to ball size/weight and release height demands.

The study highlighted IMU advantages for real-world capture but noted limitations like soft tissue artifacts and the small, non-professional sample.

Practical Applications

For NFL sports medicine and performance teams, this study underscores the value of IMU technology for baseline kinematic profiling during preseason or in-season monitoring, potentially identifying injury risks like rotator cuff tendonitis (common in pro QBs). Training protocols should emphasize hip-shoulder separation drills to enhance rotational velocity and power, integrating opposite-hip activation with shoulder strengthening for kinetic chain efficiency. Rehabilitation can target shoulder flexibility (e.g., external rotation >130°) and trunk stability to reduce overload on the throwing arm. Teams could use IMUs for individualized throw analysis, tracking changes over time to optimize mechanics, prevent overuse injuries, and inform return-to-play decisions—bridging the gap between lab data and on-field performance. Further validation with pro-level QBs is recommended to refine these applications.

Navigating the risks beyond the label: Unpacking global nutritional supplement safety. International Journal of Sports Nutrition and Exercise Metabolism, Ahead of Print: 1-33, 2025.

Purpose of the Study

This review article aims to provide a comprehensive global overview of nutritional supplement use among elite athletes, including patterns of consumption, regional regulatory frameworks, legislation, and safety measures. It synthesizes data from peer-reviewed studies, governmental sources, and expert insights across six regions (Africa, Asia, Australia/New Zealand, Europe, Latin America, and North America). A key focus is on third-party testing (TPT) programs as a strategy to mitigate risks of contamination and inadvertent anti-doping rule violations, while highlighting barriers to safe

supplement use. The goal is to inform education, policy, and practices to protect athlete health and integrity, particularly in the context of international travel and competition.

Key Findings

- **Supplement Use Patterns**: Reported use among elite athletes varies widely (7%–100%), influenced by sport, athlete level, and regional factors. Common supplements include protein powders, vitamins/minerals, creatine, caffeine, and carbohydrate-based sports drinks. High-risk products (e.g., preworkout blends, herbal formulas, weight-loss aids, proprietary blends, and certain amino acid supplements) are prevalent across all regions but to a lesser extent than basics; these are more prone to contamination with banned substances, leading to potential anti-doping violations.
- **Regional Variations**: Use is highest in regions like Europe, North America, and Australia/New Zealand (often 80%+), with lower rates in parts of Africa and Asia. High-risk supplements appear in studies from all areas, but awareness and access to safer options differ. For example, traditional herbs are noted in Africa and Asia, while preworkouts and fat burners are common globally.
- **Regulatory and Safety Frameworks**: Regulations are inconsistent worldwide. Stronger systems exist in Europe (e.g., harmonized EU standards), North America (e.g., FDA oversight in the US), and Australia/New Zealand (e.g., Therapeutic Goods Administration). In contrast, Africa, Asia, and Latin America often lack robust enforcement, leading to fragmented or outdated policies. TPT programs (e.g., NSF Certified for Sport, Informed Sport) are concentrated in the "global northwest" (Europe, North America, Australia/New Zealand), with limited availability elsewhere due to cost, language barriers, infrastructure gaps, and cultural factors.
- **Risks and Barriers**: 14%–50% of tested supplements contain prohibited substances, per analytical studies. Athlete adherence to TPT is inconsistent even where available, due to misconceptions about safety, low awareness, and financial constraints. Anecdotes highlight confusion among athletes and support staff about certification credibility and the need for batch-specific testing.
- **Education and Perceptions**: Educational efforts vary, with structured programs in wealthier regions (e.g., integrated into anti-doping curricula) but reliance on workshops elsewhere. Many athletes view supplements as low-risk, yet global heterogeneity in access and knowledge increases vulnerability during international events.

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Practical Applications

For sports medicine and performance professionals in the NFL, where players face strict anti-doping policies under the NFL-NFLPA agreement (including suspensions for PED violations), this study underscores the need for proactive risk management:

- **Adopt a "Food-First, But Not Food-Only" Approach**: Prioritize whole-food nutrition for NFL players, but when supplements are warranted (e.g., for recovery or nutrient gaps during intense training/season), guide selections based on evidence and safety. Screen for high-risk categories like preworkouts or herbals, which could trigger positives for stimulants or anabolics.
- **Prioritize TPT-Certified Products**: Recommend only batch-tested supplements from reputable programs (e.g., NSF Certified for Sport, which aligns with NFL guidelines). Implement team protocols for verifying certifications via apps or databases to minimize contamination risks—especially for players traveling internationally or sourcing from unfamiliar markets.
- **Enhance Education and Shared Responsibility**: Develop tailored education for players, coaches, and staff on global risks, using tools like the study's regional insights. Conduct regular sessions on identifying TPT labels and avoiding unverified products. Foster a team culture where supplement decisions involve nutritionists or physicians, reducing individual athlete burden.
- **Policy and Screening Integration**: Advocate for NFL-wide harmonization with global standards, such as expanding access to affordable TPT options. Incorporate supplement audits into medical evaluations, particularly for rookies or international signees, to address cultural/language barriers. Monitor for behavioral factors (e.g., misconceptions) through surveys, and collaborate with leagues like the NFL to push for better industry regulation.
- **Risk Mitigation During Travel/Off-Season**: Advise caution with foreign-sourced supplements; encourage carrying TPT-verified items or consulting ahead for equivalents. This can prevent issues like those seen in global studies, where inconsistent regulations lead to higher contamination rates.

Overall, emphasizing TPT and education can help NFL teams safeguard player eligibility, health, and performance while navigating the global supplement landscape.

Pro Football Sports Science Research Update

Optimizing agility training in team sport players – The role of perception-action coupling: A systematic review with multi-level meta-analysis. Journal of Sports Science and Medicine, 25: 84-111, 2026.

Purpose of the Study

This systematic review and multi-level meta-analysis aimed to evaluate the impact of agility-specific training on team-sport athletes, with a strong emphasis on perception-action coupling (the integration of perceptual-cognitive processes like anticipation and decision-making with physical movement). The study had three primary objectives:

1. Quantify the overall effects of such training on reactive agility performance (RAT), pre-planned agility (PAT), and reaction time (RT).
2. Identify moderating factors, including athlete characteristics (e.g., age, sex, competitive level, sport type), training modalities (e.g., duration, frequency, session length), and stimulus types (e.g., visual, visual-auditory, human).
3. Develop dose-response models to determine optimal training parameters for maximizing agility gains while minimizing risks like overtraining.

The review followed PRISMA 2020 guidelines, included 28 studies (26 for quantitative analysis), and focused on team-sport players to address gaps in prior research, which often conflated agility with change-of-direction (COD) tasks lacking perceptual elements.

Key Findings

The analysis synthesized data from 755 participants across sports like soccer, basketball, rugby, and others, though soccer and basketball dominated (limiting generalizability). Key outcomes included:

- **Overall Training Effects**: Agility training yielded a large improvement in RAT performance (Hedge's g ES = 0.65, $p < 0.01$), which requires real-time response to stimuli; a moderate improvement in PAT (ES = 0.55, $p < 0.01$), involving pre-planned movements; and a non-significant moderate effect on RT (ES = 0.52, $p > 0.05$).
- **Moderators**:
    - **Athlete Characteristics**: Junior athletes (11-14 years) showed greater gains (ES = 0.77, $p < 0.05$), as did national-level athletes (ES = 0.80, $p < 0.05$). Seniors (>18 years) and

international-level athletes had smaller but still positive effects. Females and mixed-gender groups were underrepresented, with only limited data suggesting similar gains to males.

- **Stimulus Type**: Human stimuli (e.g., opponent movements) produced large effects (ES = 0.68), followed by visual-auditory (ES = 0.83) and visual (ES = 0.57), though results were inconsistent and require caution.
- **Heterogeneity and Bias**: High within-study variability ($I^2 > 75\%$), moderate-to-low evidence certainty (per GRADE), and potential publication bias were noted. Two studies were excluded as outliers.

- **Dose-Response Relationships**:
  - Training duration showed a curvilinear pattern for RAT, peaking at 7-8 weeks (QM = 11.64, $p < 0.01$), with diminishing returns beyond 8-10 weeks.
  - Frequency had a positive linear association ($\beta = 0.172$, $p < 0.01$), favoring higher frequencies (>3 sessions/week).
  - Session length showed no significant linear relationship, but most positive effects clustered around 20-25 minutes.

Findings were robust to sensitivity analyses but highlighted the need for more diverse samples (e.g., more females, sports beyond soccer/basketball).

Practical Applications for NFL Teams

For sports medicine and performance professionals in the NFL, where reactive agility is critical for evading tackles, route-running, and defensive reads, this review underscores the value of incorporating perception-action coupling into training—beyond traditional COD drills. Practical recommendations include:

- **Program Design**: Implement 7-8 week agility blocks during offseason or in-season maintenance, with 3-5 sessions per week lasting 20-25 minutes. Integrate perceptual stimuli (e.g., human opponents or video-based simulations) to mimic game scenarios, as these may enhance transfer to on-field performance more than light-based drills. Avoid extending programs beyond 8-10 weeks without periodization to prevent plateaus or overtraining.

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- **Tailoring to Athletes**: Prioritize junior (UDFA or developmental) and national-level (e.g., practice squad) players, who show the largest gains—ideal for scouting combines or rookie camps. For veterans or elite starters, combine with sport-specific elements like position drills to maintain gains. Monitor females closely due to limited data, potentially adapting for biomechanical differences.
- **Integration and Monitoring**: Blend agility work into small-sided games or dual-task drills (e.g., reacting to a QB's throw while changing direction) to preserve perception-action links. Use reliable RAT assessments (e.g., Y-shaped or universal agility tests) pre- and post-intervention, ensuring familiarization to minimize bias. Track training load via heart rate or RPE, as volume/intensity reporting was inconsistent in studies.
- **Cautions**: Results are soccer/basketball-heavy, so validate with football-specific pilots (e.g., rugby parallels apply). Account for moderate evidence certainty and high variability—individualize based on injury history, position (e.g., WR/DB vs. linemen), and recovery metrics to avoid risks in high-stakes NFL environments. Future NFL research could compare elite vs. non-elite responses to refine these guidelines.

Optimizing NFL draft selections with machine learning classification. AI, 6: 221, 2025.

Purpose of the Study

The study aimed to address the subjectivity and high error rates in traditional NFL draft selections, which often rely on human intuition and result in significant "bust" rates (e.g., only 30% of first-round picks from 2010-2017 signed second contracts with their drafting team). Using machine learning, particularly Random Forest models, the researchers developed a system to predict player draft positions and simulate entire drafts. This involved comparing two strategies: best-player-available (BPA), which prioritizes overall talent, and need-based drafting, which incorporates team-specific positional needs. The goal was to create a data-driven approach that enhances objectivity, consistency, and predictive accuracy, ultimately helping teams avoid costly mistakes and build more competitive rosters. Data focused on post-2011 drafts (training on 2011-2019, testing on 2020 and simulating 2017), emphasizing factors like player grades, rankings, positions, and scouting text.

Key Findings

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- **Model Performance**: The Random Forest regressor achieved strong predictive accuracy for individual draft positions, with a root mean squared percentage error (RMSPE) of ~0.41 and Pearson's correlation coefficient (PCC) of ~0.94 across tests. It outperformed alternatives like Decision Tree and Support Vector Machine models. Adding physical attributes (height/weight) or text-based sentiment from scouting reports did not improve results, suggesting these are less relevant or too position-specific for generalized predictions.
- **Draft Simulation Strategies**: In simulating the 2017 first round, the BPA approach yielded 12.5% exact matches (same slot and team) and many predictions within 10 slots of actual picks. The need-based strategy performed better, with a mean absolute error (MAE) of 6.7 (vs. 8.8 for BPA) and Top-3 accuracy of 77.5% (vs. 71.9%), indicating closer alignment with real drafts. Exact matches included high-impact players like Myles Garrett and Christian McCaffrey, implying the model excels at identifying successful prospects.
- **Strategy Comparison**: Need-based drafting more realistically emulated actual NFL decisions by balancing talent with team needs, reducing deviations from real outcomes. However, neither fully captured trades or intangible factors, and the study noted AI's potential to outperform human drafts given historical bust rates.

Practical Application

For sports medicine and performance professionals in NFL teams, this AI-driven framework offers a tool to enhance pre-draft evaluations by quantifying player fit beyond subjective scouting. It can simulate scenarios to predict prospect availability, helping identify athletes with better physical potential, lower injury risk profiles (via integrated metrics like college performance), and optimal team-scheme alignment—potentially reducing bust rates and improving long-term roster health. Teams could use it to prioritize need-based picks, minimizing overreaches on "boom-or-bust" players and supporting data-informed decisions on conditioning, rehabilitation, and performance optimization. The model's quick runtime (<10 seconds per simulation) makes it feasible for real-time draft-room integration, aiding in risk mitigation and franchise profitability through sustained on-field success. Future expansions could incorporate injury history or biomechanical data for even more tailored insights.

Pro Football Sports Science Research Update

Performance-enhancing effects of inhaled medications: Implications for heart, muscle function, and doping detection in athletes. Journal of Functional Morphology and Kinesiology, 10: 462, 2025.

Purpose of the Study

This review article examines the performance-enhancing effects of inhaled medications, such as bronchodilators (e.g., β2-agonists like salbutamol, formoterol, and terbutaline) and corticosteroids, commonly used for asthma and exercise-induced bronchoconstriction (EIB) in athletes. It explores their mechanisms on the cardiovascular and muscular systems, summarizes high-profile doping cases, and evaluates detection methods. The goal is to clarify the line between legitimate therapeutic use and potential misuse, addressing challenges in maintaining fair competition while protecting athlete health. This is particularly relevant for high-contact, high-intensity sports like NFL football, where respiratory conditions are prevalent due to environmental factors and training demands.

Key Findings

- **Respiratory and Performance Benefits:** Inhaled β2-agonists primarily induce bronchodilation via cAMP-mediated smooth muscle relaxation, improving airflow, oxygen delivery, and reducing EIB symptoms. In athletes with asthma (prevalence up to 50% in endurance sports, potentially relevant for NFL conditioning), they enhance FEV1 and PEF, delaying fatigue during aerobic activities. However, evidence for ergogenic effects in healthy athletes is mixed—at therapeutic doses, benefits are minimal, but supratherapeutic use can boost sprint performance, muscle strength, and anaerobic capacity through direct effects on skeletal muscle (e.g., increased Ca2+ release, Na+/K+-ATPase activity, and glycolysis). Inhaled corticosteroids (ICS) reduce airway inflammation via glucocorticoid receptor pathways but show no significant performance gains in non-asthmatics.

- **Cardiovascular Effects and Risks:** β2-agonists activate cardiac β2-receptors, increasing heart rate, cardiac output, LVEF, and stroke volume (e.g., salbutamol improves GLS in endurance athletes, more pronounced in females). Chronic use may lower CV event rates in asthmatics due to anti-inflammatory effects, but misuse risks tachycardia, arrhythmias, hypokalemia, QTc prolongation, and autonomic imbalance. Anticholinergics (e.g., tiotropium) and theophylline pose similar risks, though less studied in athletes. ICS at high doses may contribute to hypertension and dyslipidemia long-term. Table 1 in the article summarizes these effects,

highlighting increased workload and arrhythmia risks relevant for NFL players under physical stress.

- **Muscle Function Effects:** β2-agonists promote muscle hypertrophy and contractility via PKA/Akt/mTOR pathways, favoring fast-twitch fibers (MHC IIa) and enhancing peak power during sprints or resistance efforts. Oral forms show stronger effects, but high-dose inhaled terbutaline/formoterol can improve lean mass and sprint ability in trained individuals. Benefits vary by training type (present in resistance/habitual, absent in endurance) and are dose-dependent. ICS have negligible muscle impacts. Table 2 details these outcomes, noting potential advantages for strength/power athletes but impairment with chronic terbutaline use.
- **Doping and Detection Challenges:** WADA permits specific doses (e.g., salbutamol ≤1600 µg/24h) with urinary thresholds (e.g., 1000 ng/mL for salbutamol) and TUEs for exceeding limits. Cases like Chris Froome (salbutamol, 2017) and Alberto Contador (clenbuterol, 2010) illustrate interpretation difficulties. Detection relies on urine/blood testing, but short half-lives, exercise/dehydration variability, and enantiomeric differences complicate results. Advanced methods (HRMS, chiral separation, metabolomics) improve accuracy, but false positives/negatives persist. Ethical issues include stigma, undertreatment, and diagnostic inflation for competitive edges.

Practical Application for NFL Sports Medicine and Performance Professionals

For NFL teams managing players with asthma/EIB (higher prevalence in high-intensity/contact sports), prioritize therapeutic adherence to WADA/NFL policies to avoid sanctions—monitor doses via TUEs and educate on risks like CV strain during games or training. In strength/power positions (e.g., linemen, receivers), screen for misuse, as β2-agonists could enhance sprint/explosive performance but increase injury risks (e.g., arrhythmias under tackle stress). Use tools like the Athlete Biological Passport for longitudinal monitoring, and integrate education on environmental triggers (e.g., cold/pollutants) to reduce reliance on inhalers. Balance treatment with performance optimization: therapeutic doses support recovery/endurance without ergogenic boosts, but supratherapeutic use risks health and integrity—collaborate with cardiologists for CV screening in at-risk players. Future refined detection (e.g., DBS/microsampling) could aid internal compliance testing.

Pro Football Sports Science Research Update

Practices and perceptions in hamstring training for injury prevention and enhancement of athletic performance: A qualitative analysis. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2025.

Purpose of the Study

This qualitative study aimed to explore the nuanced practices and perceptions of strength and conditioning (S&C) coaches and physiotherapists regarding hamstring training for injury prevention (specifically hamstring strain injuries, or HSIs) and athletic performance enhancement. Through semi-structured interviews with 12 practitioners (primarily from soccer, rugby, and lacrosse at youth and professional levels), the researchers sought to bridge the gap between research evidence and real-world application. The focus was on understanding challenges in programming resistance training, high-speed running (HSR), athlete monitoring, multidisciplinary team (MDT) dynamics, and translation of evidence into practice. This builds on prior survey-based work by the authors, emphasizing contextual factors like fixture congestion and individual biases that surveys cannot capture.

Key Findings

Thematic analysis revealed five main themes: areas for professional development, athlete testing/monitoring/profiling, athlete values, job role challenges, and programming. Key insights include:

- **Professional Development and Research Translation**: Practitioners expressed a need for better understanding of dose-response relationships in training (e.g., minimal effective doses for adaptations) and individualization in group settings. They incorporated recent evidence on low-volume Nordic hamstring exercises (NHEs) to improve compliance and reduce soreness, but noted gaps in research on hip-hinge exercises (e.g., Romanian deadlifts, RDLs) and combined resistance/HSR protocols. Some questioned arbitrary benchmarks like NHE force thresholds (e.g., 337 Newtons), favoring relative metrics (e.g., 10-12 Newtons per kg body mass) and session-to-session changes (5-7% variation as acceptable).

- **Athlete Testing and Monitoring**: GPS was commonly used for real-time HSR and sprint volume tracking, with interventions (e.g., 1-2 top-up reps) for athletes not meeting thresholds during games. NHE testing via NordBord was prevalent, but practitioners criticized absolute velocity thresholds, advocating for individualized relative metrics. Monitoring focused on weekly totals rather than daily spikes, with position-specific adjustments (e.g., higher HSR for faster players).

- **Athlete Values**: Education was emphasized to build buy-in, such as explaining exercise rationale (e.g., RDLs target hamstrings, not arms) and using internal cues like "feel." Muscle soreness was not a major barrier; athletes rarely resisted NHEs, and low volumes minimized delayed onset muscle soreness (DOMS). Feedback loops (e.g., daily GPS data) enhanced engagement.
- **Job Role Challenges**: Fixture congestion (e.g., midweek games) and limited athlete access were major hurdles, compounded by MDT dynamics. Effective MDTs involved collaborative planning (e.g., 4-week drill forecasts), while poor communication hindered load management. Strategies included positioning gym sessions away from games and prioritizing recovery.
- **Programming**: Resistance training emphasized movement complexity/velocity over heavy loads (e.g., self-selected weights at 60-70% body mass for youth). NHEs and RDLs were staples, with micro-dosing (e.g., 1 set of 3 reps spread across the week) during congestion to maintain adaptations without fatigue. HSR was mostly game-derived, with top-ups for low-exposure athletes; sessions incorporated competition (e.g., races) for maximal intent. Isometrics replaced eccentrics during fatigue periods to avoid de-training effects.

Overall, practices were eclectic, combining resistance and HSR, but lacked consistent progressive overload at senior levels. HSIs were seen as somewhat unavoidable, with calls for more ecologically valid research.

Practical Applications

For NFL sports medicine and performance professionals, this study underscores adaptable strategies amid similar HSI burdens (e.g., ~20-25% of NFL injuries involve hamstrings, per epidemiological data). Prioritize micro-dosing NHEs (e.g., low-rep sets 2-3x/week) during high-load periods like training camp or back-to-back games to sustain eccentric strength and fascicle length without exacerbating fatigue—potentially reducing the 15-20% recurrence rate in the NFL. Integrate GPS for individualized HSR monitoring, adjusting thresholds based on position (e.g., WRs/DBs may need more sprint top-ups than linemen), and use relative benchmarks to avoid one-size-fits-all pitfalls. Foster MDT collaboration (e.g., weekly meetings with coaches/trainers) for load planning, especially around bye weeks or playoffs, to minimize spikes. Educate players on exercise intent to boost compliance, and explore combined protocols (e.g., NHE + curved sprints mimicking routes) for performance gains. Future NFL-specific research could validate minimal maintenance doses during recovery weeks to prevent de-training, aligning with the league's shorter but intense schedule.

Pro Football Sports Science Research Update

Relationship between trunk and pelvic rotation angles during throwing and the Kerlan-Jobe Orthopedic Clinic score in American football quarterbacks. JSES International, Ahead of Print: 1-24, 2026.

Purpose of the Study

The study aimed to examine the associations between trunk and pelvic rotational movements during the throwing motion in American football quarterbacks (QBs) and their self-perceived shoulder and elbow function, as measured by the Kerlan-Jobe Orthopedic Clinic (KJOC) score. Specifically, it focused on kinematic variables at key phases—stride foot contact (SFC), maximum shoulder external rotation (MER), and ball release (BR)—during 20- and 30-yard throws, hypothesizing that these rotational mechanics would correlate with KJOC scores to provide insights into injury risk and performance.

Key Findings

- **Participant Profile and Methodology**: Involved 11 healthy collegiate or club-level QBs (mean age 20.8 years, experience 8.5 years) with no recent shoulder/elbow issues. Throws were analyzed using PitchAI (a markerless motion capture app) to measure absolute trunk rotation, pelvic rotation, and hip-shoulder separation angles. KJOC scores (mean total 89.5/100) assessed subjective function across 10 items.
- **Correlations for 20-Yard Throws**:
  - Trunk rotation at MER positively correlated with KJOC Item 3 ($r = 0.709$, $p = 0.015$; moderate-strong) and Item 7 ($r = 0.662$, $p = 0.026$; moderate).
- **Correlations for 30-Yard Throws**:
  - Hip-shoulder separation at SFC positively correlated with KJOC Item 4 ($r = 0.662$, $p = 0.026$; moderate) and strongly with pelvic rotation and hip-shoulder separation at BR ($r = 0.760$, $p = 0.007$ for Item 4).
  - Pelvic rotation at BR showed positive moderate correlations with KJOC Items 4 ($r = 0.760$, $p = 0.007$) and 10 ($r = 0.616$, $p = 0.044$).
  - Trunk rotation at MER and hip-shoulder separation at BR also showed positive correlations with various KJOC items (e.g., $r = 0.687$ to 0.734, $p < 0.05$).

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- **Overall Trends**: Greater trunk and pelvic rotations, along with increased hip-shoulder separation (indicating better energy transfer from lower body), were linked to higher KJOC scores, particularly items related to subjective performance, stability, and function during competition. No significant correlations were found for some variables, and the study noted moderate statistical power due to small sample size.

Practical Application

For NFL sports medicine and performance staff, this research highlights the value of biomechanical analysis in QB training and injury prevention. Incorporate markerless tools like PitchAI during offseason or in-season assessments to monitor rotational mechanics—focusing on enhancing hip-shoulder separation at SFC and trunk rotation at MER—to potentially reduce overload on the shoulder and elbow, which could lower injury rates (e.g., UCL tears or impingement). In rehabilitation, target drills that improve early trunk rotation and pelvic stability to boost KJOC scores and accelerate return-to-play. This could inform individualized throwing programs, especially for QBs with lower self-perceived function, and integrate with load monitoring to optimize longevity and on-field performance. Larger-scale studies in pro athletes are recommended to validate these thresholds for team protocols.

Slaves to (GPS) norms. Sport Performance & Science Reports, #274, 2025.

Purpose of the Study

This article critiques the widespread reliance on GPS-derived normative data (e.g., benchmarks, z-scores, and traffic-light systems) for managing training loads in team sports. The authors argue that these norms—often used to guide progression, avoid abrupt load changes, and minimize injury risk—lack empirical grounding as "optimal" targets. Instead, they primarily reflect typical historical practices rather than evidence-based doses that maximize performance adaptations or reduce long-term injury. The paper highlights how GPS metrics can misrepresent true neuromechanical and metabolic loads, especially in multidirectional sports, and calls for a shift back to foundational training principles (e.g., Sport Science 3.0 approach). While rooted in soccer examples, the concepts apply to high-demand contact sports like American football, where GPS is commonly used for load monitoring during practices and games.

Key Findings

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- **Lack of Evidence for Optimal Norms**: GPS norms (e.g., high-speed running [HSR] distances, acceleration/deceleration [A:D] counts) are descriptive of "usual" loads rather than prescriptive for optimal outcomes. There's limited evidence linking specific GPS thresholds to performance gains or injury reduction; for instance, only a 60-90% microcycle-to-match HSR ratio has shown some injury association, but broader norms may encode undertraining, overtraining, or poor load sequencing.
- **Misrepresentation of Loads**: Common speed-based GPS metrics underestimate mechanical demands in sport-specific training (SST), such as small-sided games or multidirectional drills. High-band A:D decreases with smaller pitches/teams despite rising exertion (e.g., due to insufficient space for high velocities and unaccounted centrifugal forces in curvilinear movements). Accelerometer metrics offer better insights but are underused due to complexity and noise.
- **Bias Toward Suboptimal Training**: Normative targets drive practices like "top-ups" (e.g., linear running to hit distance/HSR goals), which efficiently meet metrics but provide insufficient stimuli for adaptation. This favors low-risk, generic conditioning over challenging, context-specific work, potentially increasing long-term injury risk by prioritizing short-term "safety" over robustness.
- **Conflicts with Periodization and Context**: Rigid norms clash with intentional load variations (e.g., preseason ramps, rehab progressions, or fixture-driven fluctuations). Very high-intensity actions are inherently non-normative due to their rarity and exponential fatigue/injury impact.
- **Industry Pressures**: Focus on injury prevention (amid rising rates despite tech advances) leads to "load management" cultures that prioritize normative compliance for job security, blurring lines between normative and optimal.

Practical Applications for NFL Sports Medicine and Performance Professionals

- **Use Norms as Guides, Not Targets**: Incorporate GPS dashboards (e.g., z-scores, box plots) for efficient squad monitoring and communication with coaches, but avoid treating them as fixed constraints. Stray from norms when justified by rationale, such as building position-specific robustness (e.g., higher mechanical loads for linemen in agility drills) or periodizing for peak performance (e.g., ramping HSR in preseason despite exceeding benchmarks).
- **Prioritize Foundational Principles Over Metrics**: Start training design with dose-response questions: What qualities need improvement (e.g., explosive power for skill positions)? What parameters elicit them (e.g., sport-specific multidirectional drills vs. linear top-ups)? Integrate

GPS contextually—e.g., adjust for NFL-specific factors like play density, contact loads, or environmental variables (heat/humidity in Cincinnati summers)—rather than chasing arbitrary thresholds like "2x game demands."

- **Improve Metric Validity**: Advocate for advanced metrics (e.g., mechanical power from accelerometers) to better capture NFL demands like cutting, pivoting, and impacts, which speed-based data often underestimates. Collaborate on developing a "global" mechanical load measure to simplify analysis and reduce reliance on multiple bands.
- **Balance Injury Risk and Performance**: In high-stakes environments like the NFL (with dense schedules and high injury costs), use GPS to flag abrupt changes but emphasize long-term adaptations. For return-to-play, focus on progressive, context-driven loading (e.g., from linear runs to full-contact integration) rather than normative rehab targets that may underload athletes.
- **Foster Evidence-Based Cultures**: Audit training efficacy via physical testing and game performance, not just GPS norms. Educate stakeholders on limitations to counter pressures from executives or injury crises, promoting a "Sport Science 3.0" mindset that integrates tech with physiology for sustained player availability and peak output.

The impact of weekly acceleration and deceleration loads on neuromuscular performance in soccer: A session-to-session analysis. Journal of Sports Science and Medicine, 24: 851-860, 2025.

Purpose of the Study

This observational study investigated weekly variations in acceleration and deceleration loads during training sessions and their impact on neuromuscular performance in 39 male under-19 soccer players over two consecutive weeks. Using a repeated-measures design, it tracked physical demands via GNSS devices and assessed neuromuscular function through countermovement jump (CMJ) height, hamstring strength (HS), and delayed onset muscle soreness (DOMS) before and after sessions. The primary aim was to examine associations between accumulated acceleration/deceleration loads and session-to-session neuromuscular impairments, while secondarily comparing load variations and performance decrements across a standard microcycle (MD-5 to MD-1, where MD is match day).

Key Findings

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- **Neuromuscular Impairments**: Significant reductions in CMJ height (all days $p < 0.001$) and HS (all days $p < 0.001$) occurred post-training, with the largest declines on MD-4 (CMJ: -7.9%; HS: -7.9%) and MD-5 (CMJ: -7.8%; HS: -7.9%). Pre-training values showed progressive weekly decline from MD-1 to MD-4, with a slight rebound on MD-5. DOMS ratings peaked midweek on MD-3 (mean = 1.04; $F = 39.186$, $p < 0.001$), indicating cumulative fatigue buildup.
- **Load Variations**: Acceleration ($F = 248.121$, $p < 0.001$) and deceleration ($F = 227.853$, $p < 0.001$) counts were highest on MD-4 (acc: 44.00; dec: 43.65) and MD-5 (acc: 37.72; dec: 38.74), decreasing toward match day, aligning with periodization strategies emphasizing early-week high-intensity drills (e.g., small-sided games) and late-week tapering.
- **Associations**: No statistically significant correlations emerged between daily acceleration/deceleration volumes and acute changes in CMJ or HS (r values trivial to moderate, all $p > 0.05$). A non-significant trend toward moderate negative correlations appeared on MD-5 (acc vs. CMJ: $r = -0.30$, $p = 0.068$; acc vs. HS: $r = -0.32$, $p = 0.050$), suggesting potential late-week effects. Overall, impairments were attributed more to cumulative training effects and session timing than isolated mechanical loads.

Practical Application

For NFL sports medicine and performance staff, these findings underscore the value of monitoring acceleration/deceleration demands—common in football drills like route running, blocking, and pursuit angles—to mitigate neuromuscular fatigue. Implement routine pre- and post-session assessments (e.g., CMJ for explosive power, isometric HS tests for injury-prone hamstrings, and DOMS scales for subjective recovery) to track weekly trends, identifying mid-to-late week peaks in fatigue (analogous to MD-3 to MD-5). Structure microcycles with high-load days early (e.g., Monday-Tuesday for intense conditioning) followed by tapering (e.g., Thursday-Friday for tactical work), ensuring recovery before games to optimize readiness and reduce injury risk. Integrate GNSS tracking to quantify loads, allowing individualized adjustments for positions with high eccentric demands (e.g., WRs, DBs), potentially extending player longevity in-season.

Pro Football Sports Science Research Update

The perceptions and applications of cluster sets in strength and conditioning. International Journal of Sports Science & Coaching, Ahead or Print: 1-10, 2025.

Purpose of the Study

This survey-based study aimed to investigate strength and conditioning (S&C) professionals' perceptions of cluster sets—a resistance training structure incorporating short rests between repetitions or groups of reps to mitigate fatigue—and how they implement them in practice. A secondary goal was to evaluate the alignment between real-world programming tactics and evidence-based recommendations from scientific literature. The researchers surveyed 46 experienced S&C coaches (average 12.3 years in the field) across various sports contexts, including professional, collegiate, and amateur levels, to bridge the gap between research and applied settings. This is particularly relevant for NFL teams, where optimizing resistance training for power, strength, and fatigue management is critical amid large rosters, tight schedules, and high-stakes performance demands.

Key Findings

- **Adoption and Usage**: 80% (37/46) of respondents reported using cluster sets, with 43% (16/37) incorporating advanced variations (e.g., undulating, ascending, or descending clusters). Non-users cited supervision challenges with large groups (56%), doubts about effectiveness (44%), and increased session time (33%).
- **Programming Alignment with Phases**: Cluster sets were most commonly prescribed in strength-power phases (84%), aligning with literature recommending them for rate of force development and power output. Usage dropped in hypertrophy/strength-endurance phases (35%) and basic strength phases (49%), consistent with evidence favoring traditional sets for muscle endurance or hypertrophy.
- **Exercise Selection and Variables**: Bench press and squat variations were the top exercises (used by ~70-80% of respondents), despite potential drawbacks like re-racking demands. Key manipulations included higher intensities (2-15% above traditional sets, often based on %1RM), lower reps (≤5 in strength-power phases), and rest intervals of 11-30 seconds (inter-rep or intra-set). Programs typically ran 1-2 sessions/week for 3-4 weeks.
- **Perceptions and Concerns**: 78% (29/37) felt confident in programming cluster sets, with academic journals as the primary information source (70%). Half had no concerns, but others noted extended training time (27%) and supervision issues in group settings (24%). Respondents

desired more insights on periodization, population-specific applications (e.g., large teams), and targeted adaptations (e.g., power vs. hypertrophy).

- **Evidence Alignment**: Practices largely matched literature, such as using cluster sets for velocity maintenance and higher loads, but diverged in exercise choice (literature suggests weightlifting derivatives for easier implementation) and under-explored rest-redistribution sets as a time-efficient alternative.

Practical Applications for NFL Sports Medicine and Performance Professionals

Cluster sets offer a low-tech tool to enhance power and velocity in resistance training, ideal for NFL athletes needing explosive strength for tasks like sprinting, jumping, and tackling. In team settings with 50+ players, prioritize exercises avoiding frequent re-racking (e.g., power cleans or derivatives) to minimize fatigue and injury risk during setup—extending rest intervals if squats or bench presses are used. Integrate into offseason or in-season strength-power blocks at higher intensities (5-15% above traditional sets) to boost force development without excessive volume, potentially improving on-field metrics like 40-yard dash times or change-of-direction speed. Address logistical barriers by pairing with spotters for supervision or opting for rest-redistribution (redistributing rest without adding time) to keep sessions under 60 minutes. Monitor via velocity tracking if available, and educate athletes on benefits to ensure buy-in; future research on NFL-specific adaptations could refine these tactics for elite, high-volume environments.

The relationship between running load, strength, muscle architecture and hamstring strain injury across two seasons of elite male Australian football: A prospective cohort study. Sports Medicine – Open, 11: 146, 2025.

Purpose of the Study

This prospective cohort study aimed to examine the interactions between modifiable hamstring strain injury (HSI) risk factors—specifically running load (e.g., total weekly distance and high-speed running ≥24 km/h), eccentric knee flexor strength, and biceps femoris long head (BFlh) muscle architecture (e.g., fascicle length and pennation angle)—in elite male Australian Football League (AFL) players. The research investigated both univariate associations and multivariate models to better understand how these factors collectively influence prospective HSI risk over two seasons. While conducted in AFL, the

findings are highly relevant to NFL contexts, where players face similar demands from high-intensity running, sprinting, and sudden directional changes that stress the hamstrings.

Key Findings

- **Study Design and Participants**: Involved 299 unique elite AFL players (mean age 24 ± 4 years) across 408 player-seasons, with 67 HSIs recorded (16.4% incidence). Injuries typically occurred during high-speed running efforts, with an average return to full training in 17 ± 10 days and to matches in 23 ± 12 days.
- **Univariate Associations**: Several factors were significantly linked to higher HSI risk ($p < 0.05$):
    - Greater BFlh pennation angle (OR 1.6, 95% CI 1.2–2.2).
    - Shorter BFlh fascicle length (OR 0.6, 95% CI 0.5–0.9).
    - Higher weekly high-speed running distance (OR 1.7, 95% CI 1.2–2.4).
    - Greater weekly changes in total distance (OR 1.5, 95% CI 1.1–2.2) and high-speed running distance (OR 1.6, 95% CI 1.2–2.2).
    - Non-modifiable factors like age, height, weight, and prior ACL injury showed no significant differences, though prior HSI did ($p < 0.001$).
- **Multivariate Models**: Isolated models explained limited variance (strength/architecture: 12%; running load: 1%). The best combined model (BFlh pennation angle, fascicle length, peak eccentric force, weekly total distance, and weekly change in distance) explained 20% of HSI variance, with significant contributions from higher pennation angle (OR 1.8), shorter fascicle length (OR 0.7), lower peak force (OR 0.6), and greater change in weekly distance (OR 1.6; all $p < 0.05$). This suggests protective interactions (e.g., stronger hamstrings and longer fascicles may mitigate risks from load spikes). However, ~80% of variance remained unexplained, highlighting HSIs' multifactorial nature (e.g., unmeasured factors like running mechanics or glute activation).

Practical Applications for NFL Teams

HSIs are a top non-contact injury in the NFL, often sidelining players for weeks and impacting team performance—similar to AFL's financial and competitive costs. Integrate these insights into your programs:

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- **Screening and Monitoring**: Conduct pre-season assessments of eccentric hamstring strength (e.g., via NordBord or similar) and BFlh architecture (via ultrasound) to identify at-risk players (e.g., those with shorter fascicles or higher pennation angles). Combine with GNSS/GPS tracking for running loads, focusing on avoiding large weekly spikes in total distance or high-speed efforts (≥24 km/h threshold adaptable to NFL speeds).
- **Training Interventions**: Prioritize eccentric exercises (e.g., Nordic hamstring curls) to build strength and lengthen fascicles before camp, potentially reducing load-related risks. Periodize loads to minimize variability—e.g., gradual ramp-ups during OTAs or in-season—while ensuring recovery between high-speed drills.
- **Risk Management**: Use multivariate profiling for individualized plans; players with lower baseline strength/architecture may need conservative load progressions. Recognize limitations (e.g., 80% unexplained variance means no perfect prediction), so emphasize holistic prevention including sprint mechanics, hip/glute strengthening, and dynamic risk tracking. This could lower HSI rates, akin to AFL trends where eccentric focus has boosted overall strength levels.

The risk of all-cause injury and site-specific injury in athletes after concussion: A systematic review and meta-analysis. BMC Sports Science, Medicine, and Rehabilitation, 18(1): 30, 2025.

Purpose of the Study

This systematic review and meta-analysis aimed to quantify the increased risk of subsequent injuries in athletes following a sport-related concussion. Specifically, it focused on all-cause injuries (any new sports-related injury, excluding illnesses or non-sport conditions), as well as site-specific risks for recurrent concussions, upper extremity injuries, and lower extremity injuries. The analysis synthesized data from cohort studies across various sports and athlete levels (high school, college, professional) to inform better post-concussion management and return-to-play decisions. By following PRISMA guidelines and including 19 high-quality studies (totaling 86,879 athletes), the authors sought to address gaps in prior reviews, which often limited scope to specific sports or injury types, and to explore heterogeneity factors like follow-up duration, study design, sport type, and competition level.

Key Findings

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- **Overall Risk Elevation**: Athletes with a concussion history had nearly twice the odds of sustaining any subsequent all-cause injury (OR = 1.93; 95% CI = 1.39–2.68) compared to non-concussed peers. However, high heterogeneity ($I^2 > 90\%$) across studies means this should be viewed as an average association rather than a precise predictor for all contexts.
- **Site-Specific Risks**:
  - Recurrent concussions showed the highest risk (OR = 3.06; 95% CI = 1.81–5.17), indicating a threefold increase.
  - Upper extremity injuries were elevated (OR = 1.76; 95% CI = 1.10–2.81).
  - Lower extremity injuries had a moderate but significant increase (OR = 1.49; 95% CI = 1.06–2.09).
- **Subgroup Insights**:
  - Risk persisted and was significant beyond 6 months post-concussion (OR = 1.94), but not in shorter follow-ups, suggesting lingering subclinical deficits (e.g., in balance, proprioception, or neuromuscular control) beyond symptom resolution.
  - The association was strongest in college athletes (OR = 2.29; 95% CI = 1.53–3.44), while not statistically significant for professionals or high schoolers—potentially due to differences in reporting, training intensity, or sample sizes in professional cohorts.
  - Sport type was a significant moderator ($p = 0.038$), with higher risks in contact-heavy sports like football, though the analysis didn't isolate NFL-specific data.
- **Methodological Notes**: All included studies were high quality (NOS scores 7–9), with no major publication bias detected. Risks may be underestimated due to underreporting of concussions.

Practical Application for NFL Teams

For NFL sports medicine and performance staff, these findings reinforce the need to extend monitoring beyond clinical symptom clearance, as "clinical recovery" (resolution of self-reported symptoms and baseline exam return) does not guarantee full functional recovery—potentially leaving players vulnerable to reinjury for 6+ months. Implement enhanced return-to-play protocols incorporating objective assessments like advanced balance testing (e.g., via force plates or vestibular-ocular motor screening), neuromuscular evaluations, and targeted rehab programs focusing on proprioception and dynamic stability to address deficits in sensory-motor processing. In football-specific contexts, prioritize

recurrent concussion prevention through personalized helmet fitting, technique coaching, and graduated exposure drills. While professional-level risks weren't significant in subgroups, the overall trends suggest conservative timelines for high-impact positions (e.g., linemen, receivers), integrated with league protocols like the NFL Concussion Protocol. Track long-term injury data team-wide to identify at-risk players, and consider multidisciplinary input (e.g., neurologists, strength coaches) for individualized plans to minimize downtime and career impacts.

Timing-based strategies to minimize the impact of long-haul travel on sleep: A pilot study in elite athletes traveling for competition. Physiological Reports, 13: e70654, 2025.

Purpose of the Study

This pilot study aimed to evaluate the effectiveness of individualized timing-based interventions to mitigate sleep disruptions in elite athletes following long-haul travel across 13 time zones (from Montréal, Canada, to Beijing, China). Ten Canadian national short-track speed skaters were monitored during travel for a World Cup competition in December 2024. The intervention involved prescribing personalized bedtime schedules based on pre-season sleep habits (measured via actigraphy in June 2024), encouraging natural wake-ups (no alarms before 9 AM), and scheduling training sessions in the early afternoon (1-3 PM) to leverage zeitgebers (e.g., light, exercise, social cues) for circadian re-synchronization. Sleep was compared across phases (pre-season, baseline at home, pre-travel night, and 5 nights in Beijing) and against a control dataset from a prior study (Varesco et al., 2024) without interventions. The goal was to stabilize sleep-wake rhythms immediately upon arrival, hypothesizing improved sleep duration and quality compared to no intervention.

Key Findings

- **Sleep Stability and Duration**: Total sleep time (TST) in Beijing averaged ~493 minutes/night, not significantly different from pre-season (~485 min) or baseline at home (~453 min; $p=0.254$ and $p=0.999$, respectively), but markedly reduced pre-travel (~262 min; $p<0.001$) due to an early flight. Compared to the control dataset (no intervention), the intervention group achieved later bedtimes (~43 min later on average), later wake-ups (~73 min later), and longer TST (~51 min more overall; $p=0.0067$), with TST increasing ~10 min/night across the 5 days in Beijing (day effect: $p=0.0004$). Sleep efficiency was higher (89% vs. 82%; $p=0.0073$), driven by shorter sleep

onset latency (higher odds of immediate sleep; OR=10) and lower wake after sleep onset (WASO) normalized to TST (5.1% vs. 20%; $p<0.001$).

- **Adherence and Limitations**: Athletes adhered partially to bedtime recommendations (average 44 min earlier than prescribed), with 6/10 within 30 min. Bedtime and wake-up stabilized more than in controls (reduced daily shifts), but TST still increased gradually, indicating ongoing circadian adjustments. No napping occurred in Beijing.
- **Performance Impact**: Race rankings were unaffected by travel ($p=0.74$), with no time effect across home (Montréal) and Beijing competitions, though shorter distances (e.g., 500m) yielded better ranks ($p=0.044$). This aligns with prior findings that sleep recovers quickly (~5 days) but suggests interventions may enhance it without compromising performance.
- **Overall**: The intervention improved sleep metrics despite incomplete adherence, but athletes faced challenges resisting sleep drive upon arrival, possibly due to shared rooms, evening zeitgeber scarcity, or exercise timing.

Practical Applications for NFL Teams

NFL teams frequently face cross-country travel (e.g., East Coast to West Coast: 3 time zones), which disrupts sleep and circadian rhythms, potentially exacerbating fatigue, recovery delays, and injury risk during road games. This study's findings offer actionable strategies:

- **Personalized Sleep Scheduling**: Use actigraphy or wearables (e.g., WHOOP, Oura) during off-season or training camp to establish baseline sleep habits, then prescribe delayed bedtimes post-travel to align wake-ups with local daylight/social cues (e.g., team breakfast at 9 AM). Encourage alarm-free natural awakenings to reduce sleep debt—aim for 7-9 hours TST, as seen here.
- **Training Timing as a Zeitgeber**: Schedule post-arrival practices in the early afternoon (1-3 PM local time) to promote phase shifts via exercise, similar to light exposure. This could be integrated into travel recovery protocols, especially for West-to-East trips where phase advances are needed.
- **Implementation Tips**: Monitor adherence with actigraphy; address barriers like shared hotel rooms or high sleep drive by incorporating light exposure (e.g., morning outdoor walks) or melatonin if needed (not tested here). For NFL's shorter time shifts, combine with existing tools like caffeine timing or blue-light blocking. Test in preseason simulations to avoid competition

risks—potential benefits include ~10-50 min more sleep/night, enhancing alertness and performance without evident downsides.

- **Caveats**: Results are preliminary (small n=10 sample); larger NFL-specific studies could validate for football's physical demands. Focus on high-stakes travel (e.g., international games or playoffs) where sleep optimization directly impacts outcomes.